cuRAMSES II: The Effect of Exact Cooling on the Adaptive Mesh Refinement

Juhan Kim[1]★
[1] Korea Institute for Advanced Study (KIAS), 85 Hoegiro, Dongdaemun-gu, Seoul 02455, Republic of Korea

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ABSTRACT

The Jeans length criterion is a fundamental numerical stability requirement in self-gravitating hydrodynamic simulations: failing to resolve it by at least four cell widths produces artificial fragmentation that contaminates the star formation history. In cosmological adaptive mesh refinement (AMR) simulations, however, three distinct mechanisms conspire to violate or misapply this criterion. First, the *cosmological holdback* that prevents premature refinement at high redshift blocks the Jeans criterion along with all other refinement triggers, creating abrupt starbursts when levels are released. Second, the standard Newton–Raphson cooling solver can overshoot the thermal equilibrium temperature $T_{\rm eq}$, introducing transient errors in the Jeans length evaluation. Third, pre-computed cooling tables at a single redshift become stale as the UV background evolves. We address all three issues in the cuRAMSES code: (i) a *Jeans-independent holdback bypass* that allows the Jeans criterion to operate ahead of the holdback, with a configurable level limit to prevent runaway cascades; (ii) an *exact cooling integrator* based on the Townsend (2009) Y-function method applied to three-dimensional Grackle equilibrium cooling tables, eliminating Newton–Raphson overshooting; and (iii) *multi-redshift table interpolation* that tracks the evolving UV background across cosmic time. We quantify how cooling rate differences between the standard RAMSES solver and the Grackle library propagate into 40–80 per cent errors in the Jeans length at low densities, and demonstrate that the exact integrator removes the stochastic refinement noise caused by temperature overshooting. We further present a *refinement-only sub-grid scale turbulence model* that evolves the unresolved turbulent kinetic energy as a dynamical variable alongside the hydrodynamic quantities but restricts its role to the Jeans refinement criterion, providing a physically motivated, scale-dependent turbulent pressure floor that naturally limits the refinement depth without modifying the hydrodynamic solver or incurring any CFL penalty. Finally, we identify a secondary failure mode when the holdback is disabled — negative internal energy from the total-energy formulation in post-feedback flows — and introduce a minimal *eEOS polytropic energy floor* that prevents CFL time-step collapse without altering the thermodynamics of diffuse gas.

Key words: methods: numerical – cosmology: simulations – stars: formation – hydrodynamics – instabilities – turbulence

1 INTRODUCTION

Gravitational instability is the fundamental driver of cosmic structure formation. Jeans (1902) showed that a uniform, self-gravitating medium is unstable to perturbations on scales exceeding the Jeans length

$$\lambda_{\rm J} = \sqrt{\frac{\pi\, c_s^2}{G\,\rho}}, \tag{1}$$

where c_s is the sound speed and ρ the gas density. All scales below $\lambda_{\rm J}$ are stabilised by thermal pressure; all scales above it collapse. In astrophysical simulations of star-forming gas, the Jeans length therefore defines a *minimum resolution requirement*: if the computational grid or particle spacing cannot resolve $\lambda_{\rm J}$, the simulation produces qualitatively wrong results — artificial fragmentation that bears no relation to the true physical outcome (Truelove et al. 1997; Bate & Burkert 1997).

Cosmological simulations of galaxy formation must follow gravitational collapse across many orders of magnitude in spatial scale, from ∼ 100 Mpc cosmic web filaments down to ∼ 10 pc star-forming clouds. Adaptive mesh refinement (AMR; Berger & Oliger 1984; Berger & Colella 1989) achieves this by hierarchically subdividing cells in regions that require higher resolution, concentrating computational resources where they are needed. Major AMR codes used in cosmological contexts include RAMSES (Teyssier 2002), Enzo (Bryan et al. 2014), and ART (Kravtsov, Klypin & Khokhlov 1997). Alternative numerical techniques — smoothed particle hydrodynamics (SPH; Springel 2005), moving-mesh methods (Springel 2010), and mesh-free finite-mass schemes (Hopkins 2015) — face analogous resolution requirements (Bate & Burkert 1997; Nelson et al. 2009), but the AMR framework provides a particularly transparent mechanism for enforcing the Jeans condition: a cell is flagged for refinement whenever $\lambda_{\rm J} < N_J\,\Delta x$, where N_J is typically 4–8 and Δx is the cell width (Truelove et al. 1997).

In practice, however, enforcing the Jeans criterion in cosmological AMR simulations is complicated by three interacting effects:

(i) Cosmological holdback. To prevent premature over-refinement

★ E-mail: kjhan@kias.re.kr

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at high redshift — which would exhaust memory before the epoch of interest — codes such as RAMSES employ a *holdback* mechanism that progressively releases refinement levels as the scale factor a grows. This holdback blocks *all* refinement criteria simultaneously, including the Jeans criterion, which is a numerical *stability* requirement rather than merely a resolution preference. When a level is finally released, a burst of refinement and star formation follows (see also Dubois et al. 2014, 2021).

(ii) Cooling accuracy. The Jeans length is proportional to $T^{1/2}$, so the equilibrium temperature $T_{\rm eq}$ of the gas directly determines whether refinement is triggered. $T_{\rm eq}$ is set by the balance between radiative cooling and photo-heating by the ultraviolet (UV) background (Haardt & Madau 2012; Faucher-Giguère 2020). Errors in the cooling rate tables (Sutherland & Dopita 1993; Ferland et al. 2017), the UV background model (Courty & Alimi 2004), or the numerical integration scheme all propagate into errors in $T_{\rm eq}$ and hence in $\lambda_{\rm J}$. The standard RAMSES cooling solver uses a semi-implicit Newton–Raphson (N-R) iteration that is known to overshoot $T_{\rm eq}$ when the net cooling rate changes sign (cf. Townsend 2009), introducing stochastic noise into the Jeans length evaluation. The Grackle equilibrium chemistry library (Smith et al. 2017) provides more accurate cooling rates — including density-dependent metal cooling and non-local thermodynamic equilibrium (non-LTE) molecular weights — but the standard tabulated approach still requires a robust numerical integrator to avoid overshooting.

(iii) Redshift evolution. The metagalactic UV background evolves strongly with redshift: it rises during cosmic reionisation ($z \gtrsim 6$), peaks near $z \sim 2$–3, and declines towards $z = 0$ (Haardt & Madau 2012; Faucher-Giguère 2020). A cooling table computed at a single redshift becomes increasingly inaccurate as the simulation evolves away from that epoch, producing systematic biases in $T_{\rm eq}$ and $\lambda_{\rm J}$ that can reach factors of 2–3 during the He II reionisation epoch ($z \sim 3$–4).

Beyond thermal pressure, *turbulence* provides an additional source of pressure support that can significantly modify the effective Jeans length. Chandrasekhar (1951) and Bonazzola et al. (1987) showed that turbulent motions increase the effective sound speed from c_s to $(c_s^2 + \sigma_{\rm turb}^2)^{1/2}$, where $\sigma_{\rm turb}$ is the turbulent velocity dispersion, yielding a turbulent Jeans length

$$\lambda_{\rm J}^{\rm turb} = \sqrt{\frac{\pi\,(c_s^2 + \sigma_{\rm turb}^2)}{G\,\rho}}. \tag{2}$$

In the interstellar medium (ISM), turbulence is driven by supernova feedback, gravitational infall, magnetorotational instability, and galactic shear (Robertson & Kravtsov 2008; Kim & Ostriker 2017; Hopkins et al. 2018). In the context of star formation, Federrath & Klessen (2012) showed that turbulence simultaneously *supports* gas on large scales (raising the effective Jeans mass) and *compresses* gas locally (triggering collapse in overdense regions), so the net effect on refinement is scale-dependent.

Three broad strategies have been proposed to prevent artificial fragmentation when the thermal Jeans length falls below the grid scale:

1.0.0.1 (A) Polytropic pressure floor. The most widely adopted approach imposes a stiff equation of state $T = T_\star\,(n_{\rm H}/n_\star)^{g_\star - 1}$ above the star-formation threshold $n_\star$, typically with $g_\star = 2$ so that the Jeans length becomes independent of density (Truelove et al. 1997; Robertson & Kravtsov 2008; Schaye et al. 2015; Teyssier 2015; Hopkins et al. 2018). This prevents runaway refinement and ensures that the Truelove et al. (1997) criterion is automatically satisfied at all densities. However, the polytropic floor replaces the true thermodynamic state of the gas, artificially raising temperatures by orders of magnitude in dense star-forming regions. When $g_\star = 2$ the CFL time-step at density $n_{\rm H}$ is reduced by a factor $\sim (n_{\rm H}/n_\star)^{1/2}$, imposing a $\sim$300× penalty at $n_{\rm H} \sim 10^4\,{\rm cm}^{-3}$. More fundamentally, the polytropic floor acts uniformly across all environments and scales: gas in cold streams, the circumgalactic medium (CGM), and the low-density ISM are all subject to the same prescription, even though the physical turbulent support varies enormously between these regimes.

1.0.0.2 (B) Full sub-grid scale (SGS) dynamical models. A physically motivated alternative is to evolve the unresolved turbulent kinetic energy $e_{\rm sgs}$ as an additional conserved variable alongside the hydrodynamic quantities (Schmidt, Niemeyer & Hillebrandt 2006; Schmidt & Federrath 2011; Scannapieco & Brüggen 2008). In this framework, a turbulent pressure $P_{\rm turb} = \frac{2}{3}\rho e_{\rm sgs}$ is included in the Riemann solver alongside the thermal pressure, the effective sound speed becomes $c_{\rm eff}^2 = \gamma P/\rho + \sigma_{\rm turb}^2$, and the SGS energy evolves through pressure–dilatation (PdV) work, production by resolved shear, and dissipation at the grid scale. The appealing feature of this approach is its self-consistency: turbulent pressure enters the momentum and energy fluxes, participates in wave propagation, and modifies the Jeans criterion simultaneously.

Schmidt, Niemeyer & Hillebrandt (2006) developed the compressible SGS energy equation and calibrated the dissipation coefficient against driven turbulence box simulations, finding $C_\varepsilon \approx 1.5$. Schmidt & Federrath (2011) extended the model with a mixed closure combining eddy-viscosity and nonlinear terms, and Schmidt & Brückner (2025) provided a comprehensive review of large-eddy simulation (LES) techniques for astrophysical flows. Scannapieco & Brüggen (2008) implemented a similar buoyancy–drag SGS model in the SPH code GADGET and showed that it suppresses artificial fragmentation in galaxy mergers. Rennehan (2021) applied SGS modelling to cosmological simulations, demonstrating that it can suppress artificial fragmentation in the CGM. Semenov, Kravtsov & Gnedin (2016) used the locally evolved $\sigma_{\rm turb}$ to construct a turbulence-regulated star formation prescription that produces more realistic star formation histories than simple density-threshold models.

Despite these successes, the full SGS approach faces significant practical challenges in large-volume cosmological AMR simulations. First, the turbulent pressure increases the effective wave speed, which tightens the CFL condition by a factor $(1 + \sigma_{\rm turb}^2/c_s^2)^{1/2}$. In practice, this can reduce the time-step at fine AMR levels by a factor of 9–31× (Section 2.3), rendering the simulation prohibitively expensive. Second, the PdV coupling — in which compression does work on the turbulent pressure and vice versa — transfers energy between the resolved and SGS components at every time-step. When the SGS energy is subsequently dissipated and deposited as heat, the resulting non-physical gas heating further increases c_s and tightens the CFL condition, creating a positive feedback loop. Third, the energy cap (requiring $P_{\rm turb} \le P_{\rm therm}$) that is often imposed to prevent runaway SGS amplification is rendered ineffective by the γ-factor asymmetry: transferring SGS energy to thermal energy increases the effective sound speed because $\gamma(\gamma-1) = 10/9 > 2/3$ for $\gamma = 5/3$, so capping the SGS energy and redirecting it to thermal energy *increases* $c_{\rm eff}$ rather than decreasing it. Finally, the equilibrium between production and dissipation depends sensitively on the production mechanism: the gravitational production term $\mathcal{P}_{\rm grav} \propto \rho e_{\rm sgs}\sqrt{G\rho}$, which appears in some formulations (e.g. Scannapieco & Brüggen 2008), couples the equilibrium SGS dispersion directly to the Jeans scale ($\sigma_{\rm eq}^2 \propto G\rho\,\Delta x^2$), leading to excessive refinement suppression, while the Smagorinsky shear production (Smagorinsky 1963; Schmidt, Niemeyer & Hillebrandt 2006) provides turbulent support only in regions with strong resolved velocity gradients.

1.0.0.3 (C) Refinement-only SGS turbulent pressure floor. We propose a hybrid approach that retains the physical evolution of the SGS energy (production, advection, dissipation) but restricts its dynamical role to the refinement criterion alone. The turbulent pressure $P_{\rm turb}$ is *not* included in the Riemann solver and does not enter the CFL computation, the momentum flux, or the energy flux. It enters only the Jeans criterion (equation 2), where it raises the effective Jeans length and prevents unnecessary refinement. This decoupling eliminates the CFL penalty entirely — the hydrodynamic time-step is identical to a simulation without SGS — while still providing a physically motivated, scale-dependent turbulent pressure floor.

The key insight is that the *refinement criterion* and the *hydrodynamic solver* serve fundamentally different purposes. The refinement criterion decides where spatial resolution is needed; it should account for all sources of pressure support, including turbulence. The hydrodynamic solver advances the resolved-scale equations, where the sub-grid turbulent stress is by definition below the resolution limit and its back-reaction on the resolved flow is inherently uncertain. By keeping these two roles separate, we avoid the CFL penalty and the energy-transfer complications of the full SGS approach while providing refinement control that responds to the local dynamical state.

This approach is conceptually analogous to the polytropic floor (approach A), but with a crucial difference: the turbulent pressure is not a prescribed function of density but a *dynamically evolved* quantity that responds to the local shear, density, and flow history. The SGS energy is advected with the flow, produced by resolved strain, and dissipated on the grid-crossing time-scale, so it self-consistently adapts to different environments — from quiescent filaments to violently star-forming galaxies — without any environment-specific tuning.

Large-volume cosmological simulations such as Horizon-AGN (Dubois et al. 2014), EAGLE (Schaye et al. 2015), IllustrisTNG (Pillepich et al. 2018), NewHorizon (Dubois et al. 2021), and Horizon Run 5 (HR5; Lee et al. 2021) rely on accurate Jeans length resolution to produce converged star formation rates and galaxy properties. Any systematic error in the Jeans criterion — whether from the holdback, cooling inaccuracies, or the neglect of turbulent pressure — propagates into the predicted stellar masses, morphologies, and feedback histories.

In this paper we present four improvements to the handling of the Jeans criterion in cosmological AMR simulations, implemented in cuRAMSES:

(i) An *improved cooling module* that combines 3D Grackle equilibrium cooling tables with the Townsend (2009) exact integration scheme and multi-redshift interpolation, eliminating the N-R overshooting that contaminates the Jeans length near thermal equilibrium (Section 3).

(ii) A *Jeans-independent holdback bypass* that allows the Jeans criterion to operate ahead of the cosmological holdback, with a configurable level limit to prevent uncontrolled refinement cascades (Section 6).

(iii) Analysis of the *polytropic temperature floor* and its role in self-regulating the bypass cascade through physics-based density thresholds (Section 7).

(iv) A *refinement-only SGS turbulence model* that evolves the unresolved turbulent kinetic energy as a passive dynamical variable and uses it to compute an effective Jeans length, providing a physically motivated, scale-dependent refinement criterion without modifying the hydrodynamic solver or incurring any CFL penalty (Section 2.3).

The paper is organised as follows. Section 2 reviews AMR refinement in RAMSES, including the holdback mechanism and the refinement-only SGS turbulence model. Section 3 describes the exact cooling integrator and multi-redshift tables. Section 4 analyses the cooling rates and their impact on the Jeans length. Section 5 identifies the starburst problem caused by the holdback. Section 6 presents the Jeans bypass solution. Section 7 discusses the polytropic floor. Section 8 gives implementation details. Section 9 shows test results. Section 10 discusses implications, and Section 11 summarises our findings.

2 AMR REFINEMENT IN RAMSES

RAMSES (Teyssier 2002) is a grid-based AMR code that solves the coupled Euler–Poisson–Boltzmann equations on a fully threaded octree. Each cell at level $\ell < N_{\rm levelmax}$ is tested for refinement at every coarse time-step. The decision to subdivide a cell into $2^{N_{\rm dim}}$ children is governed by a combination of physical, geometric, and cosmological criteria.

2.1 Refinement Logic

The overall refinement flag for a cell is determined by the following Boolean expression:

$$\mathcal{R} = \neg\mathcal{H} \;\wedge\; \mathcal{G} \;\wedge\; (\mathcal{M} \vee \mathcal{S} \vee \mathcal{D} \vee \mathcal{P} \vee \mathcal{U} \vee \mathcal{J}), \tag{3}$$

where each symbol is defined below.

2.1.1 Physical Criteria (OR)

(i) **Quasi-Lagrangian mass criterion ($\mathcal{M}$).** The cell is flagged if its gas density exceeds a level-dependent threshold:

$$\rho_{\rm cell} \;\geq\; m_{\rm refine}(\ell)\,\frac{m_{\rm sph}}{V_{\rm cell}}, \tag{4}$$

where $m_{\rm sph}$ is the effective SPH kernel mass and $V_{\rm cell} = (\Delta x_\ell)^3$. This criterion ensures that regions containing roughly $m_{\rm refine}$ or more particles per cell are refined, providing a quasi-Lagrangian resolution that tracks gravitational collapse. Controlled by the namelist array `m_refine(1:nlevelmax)`.

(ii) **Sink particle forcing ($\mathcal{S}$).** If `sink_refine=.true.`, any cell containing a sink cloud particle (identified by `idp` < 0 and `tp` $= 0$) is unconditionally flagged up to $N_{\rm levelmax}$, guaranteeing maximum resolution around each accretion region.

(iii) **Density gradient ($\mathcal{D}$).** The normalised density gradient between a cell and each of its $2N_{\rm dim}$ face-neighbours is computed as

$$\epsilon_\rho = 2\,\max\!\left(\frac{|\rho_+ - \rho_0|}{|\rho_+ + \rho_0| + \rho_{\rm floor}},\; \frac{|\rho_0 - \rho_-|}{|\rho_0 + \rho_-| + \rho_{\rm floor}}\right). \tag{5}$$

If $\epsilon_\rho >$ `err_grad_d` along any axis, the cell is flagged.

(iv) **Pressure gradient ($\mathcal{P}$).** Analogous to the density gradient but applied to the thermal pressure:

$$\epsilon_P = 2\,\max\!\left(\frac{|P_+ - P_0|}{|P_+ + P_0| + P_{\rm floor}},\; \frac{|P_0 - P_-|}{|P_0 + P_-| + P_{\rm floor}}\right), \tag{6}$$

flagging when $\epsilon_P >$ `err_grad_p`.

(v) **Velocity gradient ($\mathcal{U}$).** A normalised velocity jump relative to the local sound speed:

$$\epsilon_v = 2\,\max\!\left(\frac{|v_+ - v_0|}{c_{s,+} + c_{s,0} + |v_+| + |v_0| + v_{\rm floor}},\; \frac{|v_0 - v_-|}{c_{s,0} + c_{s,-} + |v_0| + |v_-| + v_{\rm floor}}\right),$$

(7)

flagging when $\epsilon_v >$ `err_grad_u` in any component.

(vi) **Jeans length ($\mathcal{J}$).** When `jeans_refine`$(\ell) = N_J > 0$, the cell is flagged if the local Jeans length is shorter than N_J cell widths:

$$\lambda_{\rm J} = \sqrt{\frac{\pi\, c_s^2}{G\,\rho}} < N_J\, \Delta x_\ell, \tag{8}$$

implementing the Truelove et al. (1997) condition to prevent artificial fragmentation in self-gravitating gas. An additional density threshold `d_jeans_thre` restricts this criterion to cells with $n_{\rm H} \geq$ `d_jeans_thre`.

2.1.2 Geometry Mask (AND)

Ellipsoidal region ($\mathcal{G}$). When `r_refine`$(\ell) > -1$, only cells inside a generalised ellipsoidal region are permitted to refine. The test is

$$\left(\frac{2|\Delta x|}{r}\right)^{e_r} + \left(\frac{2|\Delta y|}{a\,r}\right)^{e_r} + \left(\frac{2|\Delta z|}{b\,r}\right)^{e_r} < 1, \tag{9}$$

where $r =$ `r_refine`, $a =$ `a_refine`, $b =$ `b_refine`, and $e_r =$ `exp_refine`. Cells outside this region are *un-flagged* regardless of physical criteria, restricting high resolution to a zoom-in target.

2.1.3 Cosmological Holdback (NOT)

Resolution holdback ($\mathcal{H}$). In cosmological runs with cooling, overly eager refinement at high redshift can exhaust memory before the epoch of interest. When `q_refine_holdback=.true.` and the physical cell size $\Delta x_\ell < 4^{1/N_{\rm dim}}\, \Delta x_{N_{\rm levelmax}}/a_{\rm exp}$ for levels $\ell >$ `nlevelmax_part` + 3, the *entire* level is blocked from further refinement:

$$\mathcal{H}(\ell) = \left[\ell > \ell_{\rm part} + \ell_{\rm collapse}\right] \;\wedge\; \left[\Delta x_\ell < 4^{1/N_{\rm dim}}\, \Delta x_{\rm min}/a\right], \tag{10}$$

where $\ell_{\rm part}$ is the particle initial-condition level (typically $\ell_{\rm min}$) and $\ell_{\rm collapse} = 3$. The first holdback level $\ell_{\rm hb}$ is the smallest ℓ satisfying equation (10):

$$\ell_{\rm hb} = \min\left\{\ell > \ell_{\rm part} + \ell_{\rm collapse} \;:\; \Delta x_\ell < 4^{1/N_{\rm dim}}\, \Delta x_{\rm min}/a\right\}. \tag{11}$$

2.2 Buffer Zones

After applying equation (3), flagged cells are expanded by `nexpand`(ℓ) cell widths in each direction via a dilation operator (`smooth_fine`). This cubic buffer ensures smooth refinement boundaries, preventing the Godunov solver from encountering abrupt level transitions.

2.3 Sub-Grid Scale Turbulence Model

The purely thermal Jeans length (equation 8) ignores the additional pressure support provided by unresolved turbulent motions. In cosmological simulations without a polytropic equation of state ($g_\star < 2$), the Jeans length $\lambda_{\rm J} \propto \rho^{(g_\star-2)/2}$ shrinks as density increases, triggering runaway refinement cascades that exhaust memory and inflate the number of fine time-steps by up to 7×. The *cosmological holdback* (Section **??**) was introduced as a pragmatic workaround, but it blocks all refinement criteria indiscriminately and introduces artificial starbursts when levels are released.

We address this root cause by implementing a *refinement-only sub-grid scale* (SGS) turbulence model that evolves the unresolved turbulent kinetic energy alongside the hydrodynamic variables, following the compressible SGS framework of Schmidt, Niemeyer & Hillebrandt (2006); Schmidt & Federrath (2011); Scannapieco & Brüggen (2008). The turbulent pressure enters the Jeans refinement criterion but is *not* included in the Riemann solver, providing a physically motivated turbulent pressure floor that naturally limits refinement depth without modifying the hydrodynamic solver or incurring any CFL penalty (see Section 1 for motivation).

2.3.1 Governing Equations

We introduce a new conserved variable $\rho e_{\rm sgs}$, the SGS turbulent kinetic energy density, advected alongside the standard hydrodynamic variables $(\rho, \rho\mathbf{v}, E)$. The evolution equation is

$$\frac{\partial(\rho e_{\rm sgs})}{\partial t} + \nabla\cdot(\rho e_{\rm sgs}\,\mathbf{v}) = -P_{\rm turb}\,\nabla\cdot\mathbf{v} + \mathcal{P}_{\rm grav} - \varepsilon_{\rm diss}, \tag{12}$$

where the three source terms on the right-hand side represent, respectively, the pressure–dilatation coupling, gravitational production, and dissipation at the grid scale.

The *turbulent pressure* is defined as

$$P_{\rm turb} = \tfrac{2}{3}\,\rho\, e_{\rm sgs}, \tag{13}$$

which acts as an isotropic closure of the Reynolds stress tensor $\tau_{ij}^{\rm sgs} \approx \frac{2}{3}\rho\, e_{\rm sgs}\,\delta_{ij}$ (Schmidt, Niemeyer & Hillebrandt 2006). The total effective pressure entering the momentum equation is $P_{\rm eff} = P_{\rm therm} + P_{\rm turb}$.

The *SGS turbulent velocity dispersion* is related to the specific SGS energy by

$$\sigma_{\rm turb} = \sqrt{\tfrac{2}{3}\, e_{\rm sgs}}, \tag{14}$$

so that $P_{\rm turb} = \rho\,\sigma_{\rm turb}^2$.

2.3.1.1 Pressure–dilatation (PdV) coupling. The first source term in equation (12), $-P_{\rm turb}\,\nabla\cdot\mathbf{v}$, represents the work done by turbulent pressure during compression ($\nabla\cdot\mathbf{v} < 0$) or expansion ($\nabla\cdot\mathbf{v} > 0$). In compression, $e_{\rm sgs}$ increases at the expense of the resolved kinetic energy, while in expansion it decreases. This term is active only when `sgs_hydro=.true.`, where the corresponding work is removed from the thermal/kinetic budget by the $P_{\rm turb}$ contribution to the Riemann solver fluxes (Section 2.3.3). In the default refinement-only mode (`sgs_hydro=.false.`), the PdV term is disabled because $P_{\rm turb}$ does not enter the Riemann flux, and applying PdV without the corresponding flux would create energy from nothing.

2.3.1.2 Gravitational production. In self-gravitating systems, gravitational contraction stirs turbulent motions at the grid scale. We model this as

$$\mathcal{P}_{\rm grav} = C_{\rm prod}\,\rho\, e_{\rm sgs}\,\sqrt{f_G\,\rho}, \tag{15}$$

where $C_{\rm prod}$ is a dimensionless production coefficient and f_G is the gravitational coupling factor:

$$f_G = \begin{cases} \dfrac{3\,\Omega_m}{8\pi\, a} & \text{(cosmological)}, \\ 1 & \text{(non-cosmological)}, \end{cases} \tag{16}$$

identical to the factor appearing in the Jeans length evaluation. The production rate $\sqrt{f_G\rho}$ has dimensions of inverse time and is proportional to the local free-fall rate $\sim (G\rho)^{1/2}$, ensuring that turbulent energy is injected most vigorously in collapsing regions where the Jeans criterion is most critical.

2.3.1.3 Dissipation. Turbulent energy cascades from the grid scale Δx to smaller scales where it is dissipated into heat. Following Schmidt, Niemeyer & Hillebrandt (2006); Semenov, Kravtsov & Gnedin (2016), the dissipation rate is

$$\varepsilon_{\rm diss} = C_{\rm diss}\,\rho\,\frac{\sigma_{\rm turb}^3}{\Delta x} = C_{\rm diss}\,\rho\,\frac{(2e_{\rm sgs}/3)^{3/2}}{\Delta x}, \tag{17}$$

where $C_{\rm diss}$ is a dimensionless coefficient calibrated to reproduce the Kolmogorov energy transfer rate. The $\sigma^3/\Delta x$ scaling follows directly from dimensional analysis of the turbulent cascade: the energy transfer rate per unit mass is $\sim \sigma^3/\ell$ at the driving scale ℓ.

Since $\varepsilon_{\rm diss} \propto e_{\rm sgs}^{3/2}$, this term is highly non-linear and can be stiff when $e_{\rm sgs}$ is large. The dissipation time-scale $t_{\rm diss} = e_{\rm sgs}/\varepsilon_{\rm diss} \propto \Delta x/\sigma_{\rm turb}$ can be much shorter than the hydrodynamic time-step, demanding an implicit treatment (Section 2.3.4).

In the refinement-only mode (`sgs_hydro=.false.`, the default), dissipated SGS energy is simply removed rather than deposited as heat. This avoids non-physical gas heating that would otherwise tighten the CFL condition (see Section 1). When the full hydrodynamic coupling is enabled (`sgs_hydro=.true.`), the dissipated energy is deposited as thermal energy, maintaining total energy conservation:

$$\left.\frac{\partial E_{\rm therm}}{\partial t}\right|_{\rm sgs} = +\varepsilon_{\rm diss}. \tag{18}$$

2.3.2 Equilibrium Analysis and Self-Regulation

In a quasi-static gravitationally collapsing region, the production and dissipation terms in equation (12) reach a local balance. Setting $\mathcal{P}_{\rm grav} = \varepsilon_{\rm diss}$ and solving for the equilibrium SGS velocity dispersion yields

$$\sigma_{\rm eq}^2 = \left(\frac{C_{\rm prod}}{C_{\rm diss}}\right)^2 f_G\,\rho\,\Delta x^2. \tag{19}$$

The corresponding equilibrium turbulent Jeans length is

$$\lambda_{\rm J}^{\rm eff} = \sqrt{\frac{\pi\,(c_s^2+\sigma_{\rm eq}^2)}{G\,\rho}} \geq \sqrt{\frac{\pi\,\sigma_{\rm eq}^2}{G\,\rho}} = \frac{C_{\rm prod}}{C_{\rm diss}}\sqrt{\frac{\pi f_G}{G}}\,\Delta x. \tag{20}$$

In the cosmological case where $f_G = 3\Omega_m/(8\pi a)$ and $G = 3\Omega_m H_0^2/(8\pi)$ in code units, the ratio $f_G/G = 1/(aH_0^2)$ is independent of density. Therefore the number of cells per effective Jeans length at equilibrium is

$$N_{J,\rm eff} = \frac{\lambda_{\rm J}^{\rm eff}}{\Delta x} \geq \frac{C_{\rm prod}}{C_{\rm diss}}\sqrt{\frac{\pi f_G}{G}}. \tag{21}$$

For our fiducial parameters $C_{\rm prod} = 1.0$, $C_{\rm diss} = 0.5$, this gives $N_{J,\rm eff} \approx 3.5$, close to the Truelove et al. (1997) minimum of 4. Importantly, since $\sigma_{\rm eq}^2 \propto \Delta x^2$, the turbulent pressure support automatically adjusts to the local grid scale: finer cells have less turbulent support, so they are refined only when thermally warranted. This is in sharp contrast to a uniform polytropic floor ($g_\star = 2$), which raises the sound speed uniformly across all scales and produces catastrophic CFL time-step reductions (a factor ~300 at high densities).

The self-regulation mechanism works as follows: if a cell is refined, its child cells have half the grid spacing, so $\sigma_{\rm eq}^2 \propto \Delta x^2$ drops by a factor of 4. The effective Jeans length decreases, and the Jeans criterion becomes harder to satisfy — *unless the thermal Jeans length is genuinely unresolved*. The cascade therefore terminates when the thermal Jeans length is resolved by $\sim N_{J,\rm eff}$ cells, precisely the behaviour desired.

2.3.3 Riemann Solver Modifications

When the full hydrodynamic coupling is enabled (`sgs_hydro=.true.`), the turbulent pressure $P_{\rm turb}$ enters the Riemann solver to maintain consistency between the SGS source terms and the resolved-scale fluxes. All five Riemann solvers available in RAMSES (HLLC, HLL, Lax–Friedrichs, acoustic, and Roe-approximate) are modified accordingly. In the default refinement-only mode (`sgs_hydro=.false.`), these modifications are inactive and the Riemann solver operates identically to the standard code; the SGS energy is advected as a passive scalar and the turbulent pressure enters only the Jeans criterion (Section 2.3.5).

2.3.3.1 Effective pressure and wave speeds. In each solver, the total pressure driving the momentum flux is replaced by

$$P_{\rm tot} = P_{\rm therm} + P_{\rm turb} = P_{\rm therm} + \tfrac{2}{3}\,\rho\,e_{\rm sgs}, \tag{22}$$

and the characteristic wave speed is computed from the effective sound speed:

$$c_{\rm eff}^2 = \frac{\gamma\,P_{\rm therm} + P_{\rm turb}}{\rho} = \frac{\gamma\,P_{\rm therm}}{\rho} + \sigma_{\rm turb}^2. \tag{23}$$

Note that the coefficient of $P_{\rm turb}$ in equation (23) is unity (not γ), because $P_{\rm turb}$ acts as an independent isotropic pressure with its own equation of state. The effective sound speed enters the signal speed estimates S_L, S_R (HLLC/HLL), the Lagrangian sound speed $c_L^2 = \rho^2 c_{\rm eff}^2$ (Roe-approximate), and the CFL time-step computation (`cmpdt`).

2.3.3.2 Energy flux. The total energy density $E = \frac{1}{2}\rho v^2 + e_{\rm therm}$ does *not* include $\rho e_{\rm sgs}$. Following Schmidt & Federrath (2011), the SGS energy is treated as a separate conserved quantity advected by the Godunov scheme. However, $P_{\rm turb}$ contributes to the energy flux:

$$F_E = (E + P_{\rm tot})\,v_n = (E + P_{\rm therm} + P_{\rm turb})\,v_n, \tag{24}$$

where v_n is the velocity normal to the cell interface. This ensures that the $P_{\rm turb}\,\nabla\cdot\mathbf{v}$ work done by turbulent pressure in compression/expansion is correctly accounted for in the total energy budget.

2.3.3.3 Primitive variable reconstruction. In the MUSCL–Hancock reconstruction (`ctoprim`), the sound speed used for slope limiting and characteristic tracing is likewise replaced by $c_{\rm eff}$ (equation 23), ensuring numerical stability at cell interfaces where $P_{\rm turb}$ is significant.

2.3.4 Implicit Source Term Integration

The dissipation term (equation 17) is stiff: the dissipation time-scale $t_{\rm diss} = \Delta x/\sigma_{\rm turb}$ can be orders of magnitude shorter than the CFL time-step in high-$e_{\rm sgs}$ regions. Explicit integration would require prohibitively small sub-steps.

We exploit the analytical structure of the dissipation ODE to obtain an unconditionally stable implicit solution without iteration. Consider the pure dissipation problem for $\sigma = \sqrt{2e_{\rm sgs}/3}$:

$$\frac{{\rm d}\sigma^2}{{\rm d}t} = -\frac{2}{3}\,\frac{C_{\rm diss}\,(\sigma^2)^{3/2}}{\Delta x}, \tag{25}$$

which has the exact solution

$$\sigma_{\rm new} = \frac{\sigma_{\rm old}}{1 + \dfrac{C_{\rm diss}\,\sigma_{\rm old}\,\Delta t}{2\,\Delta x}}. \tag{26}$$

This is a first-order L-stable scheme: $\sigma_{\rm new} \to 0$ as $\Delta t \to \infty$, so there is no risk of overshooting or negative energies regardless of the time-step size. The updated SGS energy is

$$e_{\rm sgs}^{\rm new} = \tfrac{3}{2}\,\sigma_{\rm new}^2, \tag{27}$$

and the dissipated energy $\Delta e_{\rm diss} = e_{\rm sgs}^{\rm old} - e_{\rm sgs}^{\rm new}$ is deposited into the thermal energy (equation 18).

The production and PdV terms are integrated explicitly (first-order) before the implicit dissipation step, using operator splitting. Since the production rate $\mathcal{P}_{\rm grav} \propto e_{\rm sgs}\sqrt{\rho}$ is at most linearly stiff, explicit integration is adequate for typical CFL time-steps.

This approach is analogous to the exact cooling integrator (Section 3): by exploiting the analytical structure of the ODE, we avoid Newton–Raphson iteration entirely and eliminate the convergence failures and overshooting that plague implicit root-finding methods applied to stiff source terms.

2.3.5 Jeans Criterion with SGS Turbulence

The Jeans length criterion (equation 8) is generalised to include turbulent pressure support:

$$\lambda_{\rm J}^{\rm eff} = \sqrt{\frac{\pi\,(c_s^2 + \sigma_{\rm turb}^2)}{G\,\rho}} < N_J\,\Delta x_\ell, \tag{28}$$

where $\sigma_{\rm turb}^2 = \frac{2}{3}e_{\rm sgs}$ is obtained from the locally evolved SGS energy. This modification appears in the `jeans_length_refine` routine and is active regardless of the `sgs_hydro` setting. In the default refinement-only mode, this is the *sole* dynamical role of the SGS energy: the Riemann solver, CFL computation, and primitive variable reconstruction operate identically to the standard code without SGS.

The combined effect provides a physically motivated turbulent pressure floor: the SGS model evolves the unresolved turbulent energy in response to the local flow, and the Jeans criterion uses this energy to (i) increase the effective Jeans length so that the AMR refinement criterion is satisfied at coarser levels, and (ii) reduce the number of fine time-steps through subcycling, improving computational efficiency — all without any time-step penalty.

2.3.5.1 Comparison with the polytropic floor. The polytropic equation of state $T = T_\star(n_{\rm H}/n_\star)^{g_\star - 1}$ serves a similar purpose — preventing unresolved fragmentation — but acts uniformly across all scales. Setting $g_\star = 2$ makes $\lambda_{\rm J} \propto \rho^0$ (constant), which prevents the cascade but imposes a CFL penalty proportional to $(n_{\rm H}/n_\star)^{1/2}$, reducing the time-step by a factor of ~300 at densities $n_{\rm H} \sim 10^4\,{\rm cm}^{-3}$. The SGS model avoids this penalty because $\sigma_{\rm eq} \propto \Delta x$ scales with the local grid spacing: fine cells have small $\sigma_{\rm turb}$ and therefore no CFL overhead, while coarse cells have large $\sigma_{\rm turb}$ that prevents unnecessary refinement.

2.3.6 Namelist Parameters

The SGS model is activated by setting `use_sgs=.true.` in `&RUN_PARAMS` and requires `NVAR=12` at compile time (the standard value is 11). The model parameters are set in the `&SGS_PARAMS` namelist block:

The default configuration uses Smagorinsky shear production ($C_{\rm smag} = 0.10$, corresponding to an eddy-viscosity coefficient $C_\nu = C_{\rm smag}\sqrt{2/3} \approx 0.082$, consistent with Schmidt, Niemeyer & Hillebrandt 2006) as the sole source of SGS energy, with $C_{\rm prod} = 0$ (gravitational production disabled). The dissipation coefficient $C_{\rm diss} = 2.76$ corresponds to $C_\varepsilon = C_{\rm diss} \times (2/3)^{3/2} \approx 1.5$ in the notation of Schmidt, Niemeyer & Hillebrandt (2006), calibrated against driven turbulence simulations. When `sgs_hydro=.false.` (the default), the turbulent pressure enters only the Jeans criterion; the Riemann solver, CFL computation, and primitive variable reconstruction are unaffected. Setting `sgs_hydro=.true.` enables the full hydrodynamic coupling described in Section 2.3.3, at the cost of a significant CFL penalty (see discussion in Section 1).

Table 1. SGS turbulence model parameters and their default values.

Parameter	Symbol	Default	Description
`sgs_C_prod`	$C_{\rm prod}$	0.0	Gravitational production coefficient
`sgs_C_diss`	$C_{\rm diss}$	2.76	Dissipation coefficient
`sgs_C_smag`	$C_{\rm smag}$	0.10	Smagorinsky shear production coefficient
`sgs_floor`	—	10^{-30}	Minimum $e_{\rm sgs}$ (positivity floor)
`sgs_cap`	—	1.0	Max ratio $P_{\rm turb}/P_{\rm therm}$
`sgs_e_init`	—	0.01	Initial $e_{\rm sgs}/e_{\rm therm}$ at restart
`sgs_hydro`	—	`.false.`	Include $P_{\rm turb}$ in Riemann solver

2.3.7 Implementation Notes

2.3.7.1 Variable infrastructure. The SGS energy density is stored as the $N_{\rm var}$-th hydro variable (index `isgs`), appended after the chemical species index `ichem` in the passive scalar chain:

`imetal` → `idelay` → `ivirial` → `ixion` → `ichem` → `isgs`.

The code verifies at startup that `isgs` ≤ `NVAR` and exits with a diagnostic message if the binary was compiled with insufficient variables.

2.3.7.2 Source term ordering. The SGS source terms are applied after cooling (`cooling_fine`) and before the next Poisson solve in the `amr_step` loop. This ordering ensures that the thermal state is updated before the SGS PdV coupling accesses the divergence field, and that the gravitational production uses the current density field.

2.3.7.3 Velocity gradients. The shear production and PdV terms require the velocity gradient tensor $\partial v_i/\partial x_j$. These are computed from face-neighbour central differences at each leaf cell, independently of the Godunov solver's `divu` array. This ensures that the SGS source terms are available regardless of the `pressure_fix` setting. From the full gradient tensor, the traceless strain-rate magnitude $|S^*|^2 = S_{ij}S_{ij} - \frac{1}{3}(\nabla\cdot\mathbf{v})^2$ and the divergence $\nabla\cdot\mathbf{v}$ are computed for shear production and PdV coupling, respectively.

2.3.7.4 Restart compatibility. When restarting from a checkpoint written with `NVAR` = 11 (no SGS), the restore routine reads 11 variables per cell and initialises the 12th (`isgs`) to zero. The SGS floor ensures positivity, and the production term rapidly builds up $e_{\rm sgs}$ from zero to its equilibrium value within a few free-fall times.

3 EXACT COOLING INTEGRATION

The cooling module determines the equilibrium temperature $T_{\rm eq}$ of the gas, which controls the Jeans length $\lambda_{\rm J} \propto \sqrt{T/\rho}$ and hence the refinement decisions. Errors in the numerical integration scheme

propagate directly into errors in λ_J and may trigger (or suppress) refinement at the wrong density and temperature. We implement an exact integration method based on the Townsend (2009) Y-function, coupled with multi-redshift table interpolation to track the evolving UV background across cosmic time.

3.1 The Townsend Exact Integration Scheme

To eliminate the numerical overshooting inherent in implicit cooling solvers (see Section 4.1.2), we implement the exact integration method of Townsend (2009), which transforms the cooling ODE into an analytically invertible form using the Y-function.

3.1.1 Y-Function Formulation

For a piecewise power-law cooling function $\Lambda(T) \propto T^{\alpha_j}$ in each temperature bin $[T_j, T_{j+1}]$, the integrated cooling variable is

$$Y(T) = \int_{T_{\rm ref}}^{T} \frac{dT'}{T'\,\hat{\Lambda}(T')} = \sum_k \Delta Y_k, \tag{29}$$

where $\hat{\Lambda} = \Lambda/\Lambda_{\rm ref}$ is the normalised cooling rate. Each segment contributes

$$\Delta Y_j = \begin{cases} \dfrac{T_j^{-\alpha_j} - T_{j+1}^{-\alpha_j}}{\alpha_j\,\hat{\Lambda}_j\,T_j^{-\alpha_j}} & (\alpha_j \neq 0), \\ \dfrac{\ln(T_j/T_{j+1})}{\hat{\Lambda}_j} & (\alpha_j = 0). \end{cases} \tag{30}$$

The exact temperature after a time-step Δt is obtained by inverting the relation

$$Y(T^{n+1}) = Y(T^n) + \Lambda_{\rm ref}\,\frac{n_{\rm H}}{E_0}\,\Delta t, \tag{31}$$

where $E_0 = (3/2)\,n_{\rm tot}\,k_{\rm B}$ is the thermal energy per unit temperature. The inversion is performed by binary search on the cumulative Y-table and local analytic inversion within the identified segment.

3.1.2 Advantages Over Newton–Raphson

The Y-function scheme has several fundamental advantages:

(i) **No overshooting.** The method integrates the cooling equation *exactly* for piecewise power-law rates. The temperature evolves monotonically towards the equilibrium; it cannot oscillate across $T_{\rm eq}$.
(ii) **No sub-stepping.** A single evaluation of equation (31) advances the solution by the full hydrodynamic time-step, regardless of $t_{\rm cool}/\Delta t$. The N-R solver requires up to 500 sub-steps when $\Delta t \gg t_{\rm cool}$.
(iii) **Correct equilibrium.** When $\Delta t \to \infty$, $Y \to Y_{\rm eq}$ and the cell relaxes exactly to $T_{\rm eq}$. The N-R solver only reaches $T_{\rm eq}$ asymptotically through iteration, and may settle on the wrong side if the sub-step structure interacts with the sign change.
(iv) **Deterministic.** The result depends only on $(T^n, n_{\rm H}, Z, \Delta t)$ and the cooling table, with no sensitivity to iteration count, convergence tolerance, or adaptive step heuristics.

The cost is a table lookup and a log/exp evaluation per cell (~few μs), comparable to a single N-R sub-step. Since the N-R solver typically requires 5–50 sub-steps, the exact scheme is faster for strongly cooling gas.

3.1.3 Implementation: Eunha Cooling Module

The exact integrator is implemented as a Fortran module (`eunha_cooling_mod.f90`) that reads a 3D Grackle equilibrium cooling table $\Lambda_{\rm net}(n_{\rm H}, Z, T)$ and pre-computes the cumulative Y-table. The module provides three runtime modes selected by the namelist parameter `cooling_method`:

- `'original'` — standard RAMSES N-R solver (default, no overhead);
- `'exact'` — Townsend exact integration replaces the N-R solver;
- `'compare'` — both solvers run; the N-R result is used for physics while per-step diagnostics report the difference $\Delta T_2^{\rm orig} - \Delta T_2^{\rm exact}$.

The Grackle table adds a third axis (metallicity Z) with 40 bins spanning 10^{-6}–$10\,Z_\odot$, compared to the 2D RAMSES table with linear Z scaling. The mean molecular weight $\mu(n_{\rm H}, Z, T)$ is also tabulated from Grackle's non-LTE equilibrium, replacing the RAMSES CIE calculation. The $T_2 \leftrightarrow T$ conversion uses an iterative scheme: $\mu \to T = T_2\,\mu \to \mu'(n_{\rm H}, Z, T) \to T' = T_2\,\mu'$, converging to < 0.1 per cent in 3 iterations.

3.2 Redshift-Dependent Cooling: Multi-z Tables

The cooling rate depends on redshift through the UV background intensity $J_\nu(z)$, which evolves as the ionising photon production rate rises during cosmic reionisation ($z \gtrsim 6$), peaks near $z \sim 2$–3, and declines towards $z = 0$ as the star formation rate density decreases (Haardt & Madau 2012; Faucher-Giguère 2020). The equilibrium temperature $T_{\rm eq}(n_{\rm H}, z)$ therefore evolves with redshift even at fixed density.

3.2.1 How the standard solver tracks redshift

The standard RAMSES cooling module calls `set_table(aexp)` at each coarse time-step, rebuilding the 2D cooling table $\Lambda(n_{\rm H}, T_2)$ using the Courty UV background model evaluated at the current expansion factor a. This analytic UV model is a polynomial fit to Cloudy 07, so the table rebuild is inexpensive and correctly tracks the redshift evolution of the background — within the limitations of the Courty model itself (Section 4.1.3).

3.2.2 The single-z limitation of the Grackle table

The Grackle-based Eunha module (`eunha_cooling_mod.f90`) initially loaded a single pre-computed 3D table at a fixed redshift (e.g. $z = 5.1$ for the HR5 restart). While the cooling rates are more accurate than the RAMSES 2D table at that redshift, the table becomes increasingly wrong as the simulation evolves away from the tabulated z. In particular:

- At $z < z_{\rm table}$, the UV background is stronger (more photoheating), so the true $T_{\rm eq}$ is higher than the tabulated value. The exact integrator would drive cells to an *artificially low* equilibrium temperature, underestimating λ_J and potentially triggering spurious refinement.
- At $z > z_{\rm table}$, the UV background is weaker, and the exact integrator would settle on a $T_{\rm eq}$ that is too high, overestimating λ_J and suppressing refinement.

- The most rapid UV evolution occurs during hydrogen reionisation ($z \sim 6$–8) and He II reionisation ($z \sim 3$–4), where $T_{\rm eq}$ can change by factors of 2–3 over $\Delta z \sim 1$.

3.2.3 *Multi-z table and runtime interpolation*

We address this by pre-computing Grackle equilibrium tables at multiple redshifts and interpolating at runtime. The offline generator (`generate_multi_z_table.py`) calls the Grackle library at each redshift snapshot to produce a single multi-z binary file containing N_z copies of the 3D table $(n_{\rm H}, Z, T)$.

The default sampling uses 10 redshift snapshots:

$$z \in \{0.0,\ 0.5,\ 1.0,\ 1.5,\ 2.0,\ 3.0,\ 4.0,\ 5.0,\ 7.0,\ 10.0\}, \tag{32}$$

with denser spacing at low z ($\Delta z = 0.5$ for $z \leq 2$) where the UV background evolves most rapidly relative to the Hubble time, and coarser spacing at high z ($\Delta z \geq 1$).

At each coarse time-step, `eunha_interp_redshift(z)` performs the following:

(i) Finds the bracketing snapshots $z_i \leq z < z_{i+1}$ and computes the linear interpolation weight $w = (z - z_i)/(z_{i+1} - z_i)$.

(ii) Interpolates the net cooling rate and mean molecular weight:

$$\Lambda_{\rm net}(n_{\rm H}, Z, T; z) = (1-w)\,\Lambda^{(i)} + w\,\Lambda^{(i+1)}, \tag{33}$$

$$\mu(n_{\rm H}, Z, T; z) = (1-w)\,\mu^{(i)} + w\,\mu^{(i+1)}. \tag{34}$$

(iii) Rebuilds the cumulative Y-table for all $N_{\rm dens} \times N_{\rm metal} = 60 \times 40 = 2400$ cells (OpenMP parallelised).

(iv) Caches the current redshift; the rebuild is skipped if $|\Delta z| < 0.01$.

The total file size for 10 snapshots on the standard $60 \times 40 \times 256$ grid is $10 \times 2 \times 614400 \times 8 \approx 94$ MB, which is distributed with the source code. Backward compatibility with single-z files is maintained through automatic format detection (the first `int32` in a multi-z file encodes N_z, which is distinguishable from the grid dimension $L = 60$).

3.2.4 *Impact on cooling overshoot*

The multi-z table eliminates a systematic error that compounds with the numerical overshoot discussed in Section 4.1.2. Consider a simulation at $z = 2$ using a table computed at $z = 5$:

(i) The UV background at $z = 2$ is $\sim$ 3–5× stronger than at $z = 5$ (Haardt & Madau 2012), so $T_{\rm eq}^{z=2} > T_{\rm eq}^{z=5}$.

(ii) A cell cooling towards equilibrium at $z = 2$ would be driven to the *wrong* (lower) $T_{\rm eq}^{z=5}$ by the stale table.

(iii) Even with the exact integrator (no numerical overshoot), the cell reaches an incorrect equilibrium, yielding a systematically too-small Jeans length.

(iv) The N-R solver compounds this: it may overshoot below the already-too-low $T_{\rm eq}^{z=5}$, further reducing $\lambda_{\rm J}$.

With the multi-z table, the Eunha module tracks the correct $T_{\rm eq}(z)$ at every coarse step, matching the RAMSES solver's `set_table(aexp)` behaviour but with the improved Grackle rates and the exact integration scheme. The combination eliminates both the systematic error (wrong UV background) and the numerical error (N-R overshoot), providing the most accurate equilibrium temperature determination available within the operator-split cooling framework.

Table 2. Test conditions for the cooling solver validation at $z = 5$. $t_{\rm cool}$ is the initial cooling time and $T_{\rm eq}$ is the equilibrium temperature from the RK4 reference.

Name	$n_{\rm H}$ [cm^{-3}]	T_0 [K]	$Z/Z_\odot$	$t_{\rm cool}$	$T_{\rm eq}$ [K]
IGM	10^{-4}	10^6	10^{-3}	1.6 Gyr	15 300
Filament	10^{-2}	10^5	10^{-2}	4.3 Myr	14 500
ISM	1	3×10^4	1	126 yr	4 020
Dense hot	10^2	10^7	1	43 kyr	171
Strong	1	10^7	1	4.3 Myr	4 020

3.3 Solver Validation

We validate the Townsend exact integrator against an adaptive fourth-order Runge–Kutta (RK4) reference solution (tolerance 10^{-8}) and the standard RAMSES Newton–Raphson (N-R) subcycling solver. Five test conditions at $z = 5$ are drawn from the multi-z Grackle table, spanning four decades in density and three decades in temperature (Table 2).

3.3.1 *Time-step independence*

Fig. 1 shows the final temperature $T_{\rm final}$ after a single cooling step as a function of $\Delta t/t_{\rm cool}$. The RK4 reference (black) and the Townsend integrator (blue) produce identical results across the full range $0.01 \leq \Delta t/t_{\rm cool} \leq 10^3$, demonstrating that the exact scheme is genuinely independent of the time-step size. The N-R solver with $\Delta_{\rm var} = 4$ (red), 20 (green), and 100 (magenta) converges to the same answer only when the time-step is short ($\Delta t \lesssim t_{\rm cool}$).

3.3.2 *Accuracy of the N-R solver*

Fig. 2 shows the relative error $|T_{\rm NR} - T_{\rm RK4}|/T_{\rm RK4}$ of the N-R solver as a function of the adaptive step parameter $\Delta_{\rm var}$, for a fixed time-step $\Delta t = 10\,t_{\rm cool}$. The ISM condition (green) requires $\Delta_{\rm var} \approx 60$ to reach 1 per cent accuracy, far exceeding the default value $\Delta_{\rm var} = 4$ (vertical dashed line). This reflects the steep cooling function near the hydrogen ionisation transition at $T \sim 10^4$–3×10^4 K, where the N-R step control $|\Delta T/T| \leq 1/\Delta_{\rm var}$ is insufficiently fine to track the rapid equilibrium approach. The Townsend integrator achieves $< 10^{-5}$ relative error for all conditions (not shown on this scale).

Fig. 3 shows the number of N-R subcycles as a function of $\Delta_{\rm var}$. The subcycle count scales as $N_{\rm sub} \sim \Delta_{\rm var} \cdot \Delta t/t_{\rm cool}$ (grey dotted lines), confirming that larger $\Delta_{\rm var}$ demands proportionally more subcycles. The Fortran default limit of `iter_max` = 500 (red dashed line) is exceeded for the strongest cooling conditions at $\Delta_{\rm var} \gtrsim 50$.

3.3.3 *Cooling trajectories*

Fig. 4 compares the temperature evolution $T(t)$ for the Strong ($n_{\rm H} = 1$, $T_0 = 10^7$ K) and ISM ($n_{\rm H} = 1$, $T_0 = 3 \times 10^4$ K) conditions over $5\,t_{\rm cool}$. The RK4 reference is plotted with fine resolution (500 steps); the Townsend and N-R solvers use coarse steps ($\Delta t = t_{\rm cool}$) to reveal the differences. In the Strong case (left), all solvers reach $T_{\rm eq} \approx 4020$ K within one cooling time. In the ISM case (right), the N-R solver with $\Delta_{\rm var} = 4$ (red dashed) systematically overshoots the reference trajectory, while the Townsend integrator (blue) tracks the RK4 curve closely even with a single step per $t_{\rm cool}$.

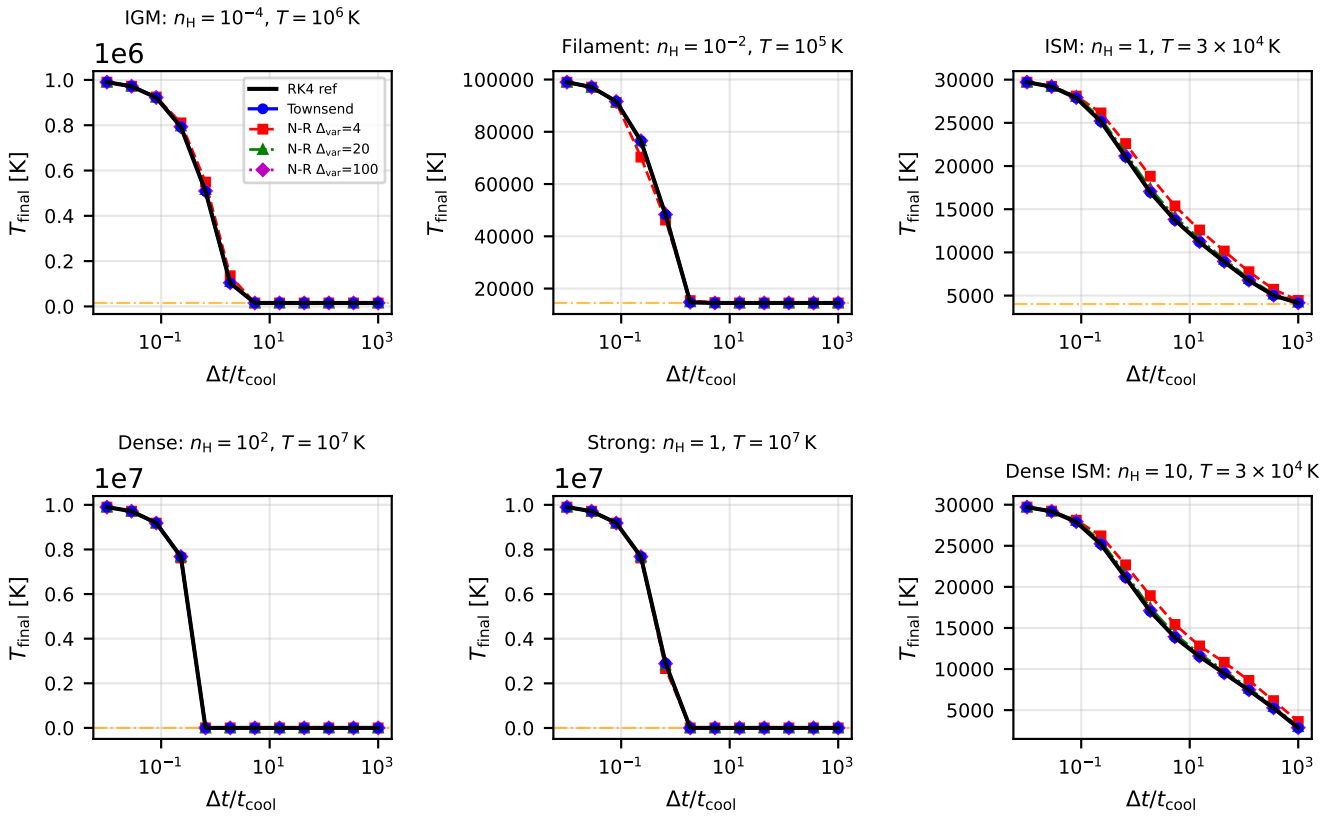

Figure 1. Final temperature after a single cooling step versus $\Delta t/t_{cool}$ for the five test conditions. The Townsend exact integrator (blue circles) overlaps the RK4 reference (black line) at all time-steps. The orange dashed line marks the equilibrium temperature T_{eq}. The N-R solver (red, green, magenta) converges only for small Δt.

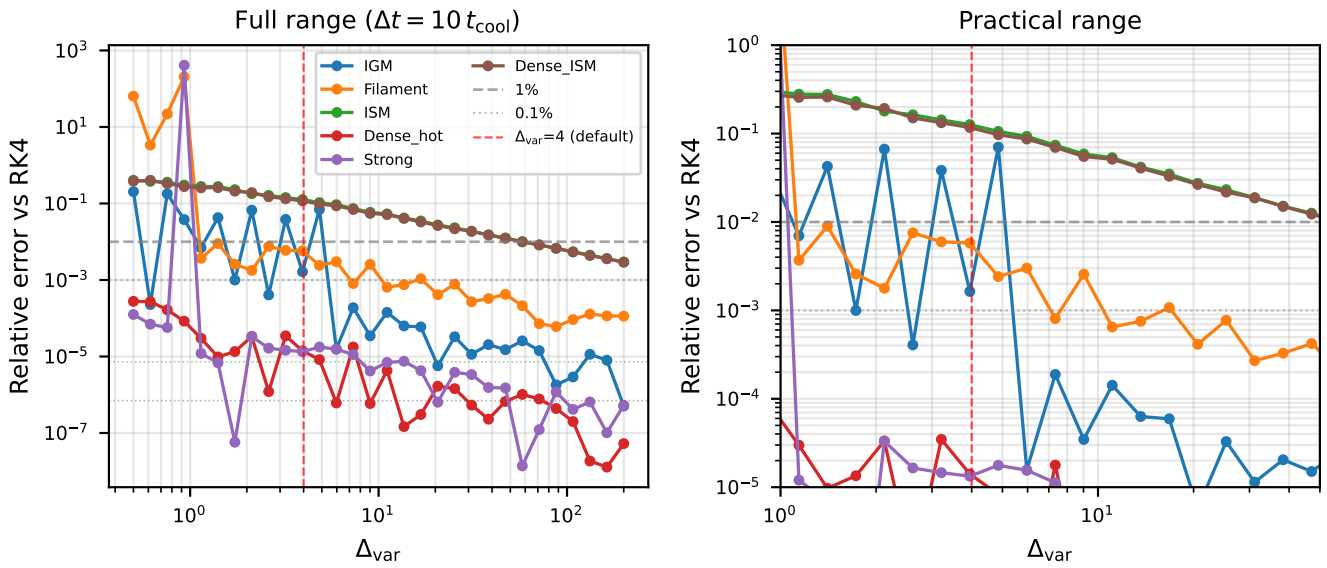

Figure 2. Relative error of the N-R solver versus Δ_{var} at $\Delta t = 10\, t_{cool}$. *Left:* full range; *right:* practical range. The grey dashed lines mark 1 per cent and 0.1 per cent thresholds. The ISM condition (green) requires $\Delta_{var} \approx 60$ for 1 per cent accuracy.

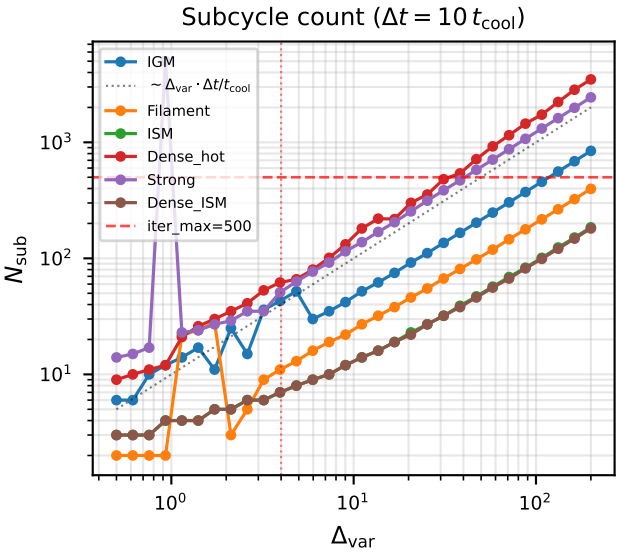

Figure 3. Number of N-R subcycles versus $\Delta_{\rm var}$ at $\Delta t = 10\,t_{\rm cool}$. Grey dotted lines show the theoretical scaling $\Delta_{\rm var} \cdot \Delta t/t_{\rm cool}$. The red dashed line marks the Fortran iteration limit `iter_max` = 500.

4 COOLING RATES AND THE JEANS LENGTH

Having described the exact integration scheme, we now analyse the cooling rates themselves. The underlying cooling tables and UV background model differ between the standard RAMSES solver and the Grackle equilibrium chemistry library (Smith et al. 2017). These differences propagate into the equilibrium temperature and hence affect the Jeans length evaluation and refinement decisions.

4.1 The Standard RAMSES Cooling Solver

The RAMSES cooling module solves the energy equation

$$\frac{dT_2}{dt} = -\frac{2X_{\rm H}}{3k_{\rm B}}\,\frac{\Lambda_{\rm net}(n_{\rm H}, T_2)}{n_{\rm H}}, \tag{35}$$

where $T_2 \equiv T/\mu$ is the temperature per mean molecular weight and $\Lambda_{\rm net}$ is the net volumetric cooling rate (cooling − heating) in erg cm^{-3} s^{-1}.

The solver uses a semi-implicit Newton–Raphson iteration. At each sub-step, the temperature is updated as

$$T_2^{(n+1)} = T_2^{(n)}\,\frac{1 + \Lambda'/w_{\rm cool} - \Lambda/(T_2^{(n)}\,w_{\rm cool})}{1 + \Lambda'/w_{\rm cool}}, \tag{36}$$

where $\Lambda' \equiv d\Lambda_{\rm net}/dT_2$ and the adaptive step control parameter is $w_{\rm cool} = \max\left(|\Lambda|\,\Delta_{\rm var}/T_2,\ w_{\rm max},\ -\Lambda'\,\Delta_{\rm var}\right)$ with $\Delta_{\rm var} = 4$.

The cooling rates $\Lambda_{\rm net}$ and their derivatives are interpolated from a pre-computed 2D table $\Lambda(n_{\rm H}, T_2)$, built at the start of the simulation for the current expansion factor a. The table contains separate contributions from:

- H/He collisional cooling (ionisation, recombination, bremsstrahlung, excitation) computed from collisional ionisation equilibrium (CIE; Sutherland & Dopita 1993);
- UV photo-heating from the Courty & Alimi (2004) background model;
- Compton cooling/heating off the CMB;
- metal cooling from the Cloudy 07 table (Plöckinger & Schaye 2015), scaled linearly with metallicity Z.

4.1.1 *Adaptive Sub-stepping (Subcycling)*

It is sometimes stated that the RAMSES cooling solver uses subcycling to prevent overcooling. This is correct: rather than applying equation (36) once per hydrodynamic time-step Δt, the solver integrates forward in time through a sequence of adaptive sub-steps. The algorithm maintains a pseudo-time variable $t_{\rm sub}$, initialised to zero, and repeats the linearised implicit update until $t_{\rm sub}$ reaches the target $t_{\rm max} = (2X_{\rm H}/3k_{\rm B})\,n_{\rm H}\,\Delta t$. Each sub-step advances $t_{\rm sub}$ by $\delta t_{\rm sub} = 1/w_{\rm cool}$, where $w_{\rm cool}$ is the adaptive rate defined in equation (36). Cells that reach $t_{\rm max}$ are deactivated; the loop terminates when all cells are done or after 500 iterations. A final linear interpolation between the last two sub-steps ensures that the solution lands exactly at $t_{\rm max}$:

$$T_2^{\rm final} = T_2^{(n)}\,\frac{t_{\rm max} - t^{(n-1)}}{t^{(n)} - t^{(n-1)}} + T_2^{(n-1)}\,\frac{t^{(n)} - t_{\rm max}}{t^{(n)} - t^{(n-1)}}. \tag{37}$$

The key parameter controlling sub-step size is $\Delta_{\rm var} = 4$, which limits the relative temperature change per sub-step to roughly a factor of 4. When the cooling rate is large, the solver takes many small sub-steps and tracks the temperature evolution accurately. In practice, however, three issues remain:

(i) **First-order accuracy.** Each sub-step is a linearised implicit Euler update — first-order in $\delta t_{\rm sub}$. Higher-order schemes (e.g. Runge–Kutta) would reduce the number of sub-steps needed for the same accuracy, but the RAMSES solver does not employ them.

(ii) **Permissive step control.** $\Delta_{\rm var} = 4$ allows the temperature to change by up to a factor of 4 in a single sub-step. Near the cooling–heating equilibrium, where the temperature needs to settle to $T_{\rm eq}$ with per cent-level accuracy (since the Jeans length scales as $T^{1/2}$), this is too coarse.

(iii) **Final interpolation error.** The linear interpolation (equation 37) assumes that T_2 varies linearly between the last two sub-steps. If the net cooling rate changes sign in this interval (i.e. the cell crosses $T_{\rm eq}$ between sub-steps $n-1$ and n), the interpolation can land on the wrong side of the equilibrium.

Thus, while subcycling mitigates the most extreme errors, it does not eliminate the systematic biases discussed below.

4.1.2 *Time-Step Dependence and Spurious Temperature Discontinuities*

The most damaging property of the N-R subcycling solver in an AMR context is that the computed final temperature depends on the hydrodynamic time-step Δt. Fig. 1 demonstrates this directly: for the ISM and Dense ISM conditions ($n_{\rm H}$ = 1–10 cm^{-3}, $T_0 = 3 \times 10^4$ K), the N-R solver with $\Delta_{\rm var} = 4$ returns systematically different temperatures as $\Delta t/t_{\rm cool}$ varies from 10^{-2} to 10^3. The Townsend exact integrator, by contrast, is genuinely Δt-independent across the entire range.

In an AMR simulation with subcycling (`nsubcycle` = 1, 1, 2), adjacent cells at different refinement levels receive time-steps differing by factors of 2. Cells at the same physical conditions but on different levels therefore receive *different* final temperatures from the N-R solver. This creates artificial temperature discontinuities at level boundaries. Through the Jeans length $\lambda_{\rm J} \propto T^{1/2}\,\rho^{-1/2}$, these temperature jumps translate directly into spurious refinement triggers: a cell on the coarser level may remain above the Jeans threshold while its

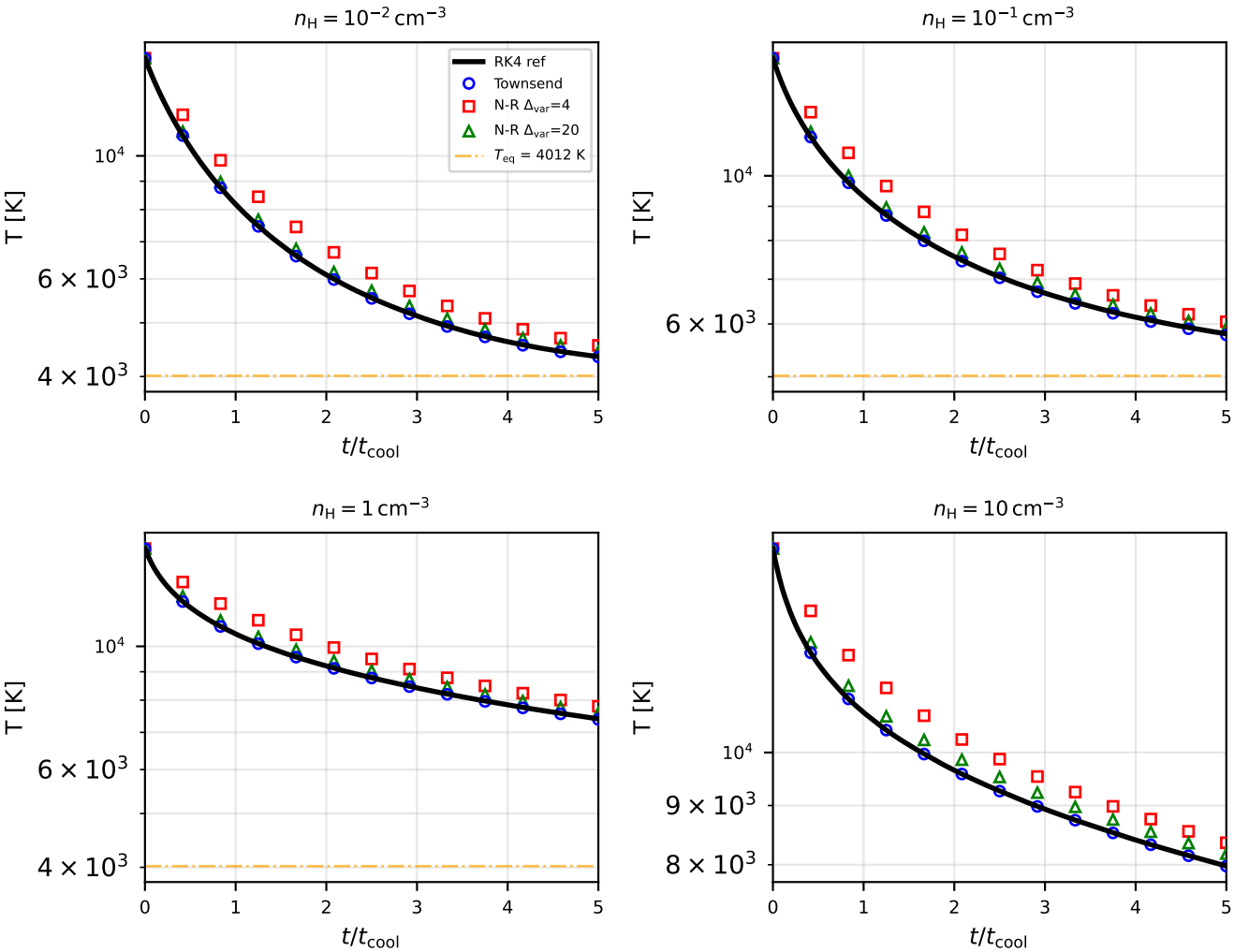

Figure 4. Temperature trajectories $T(t/t_{cool})$ for the Strong (left) and ISM (right) conditions. The RK4 reference (black solid, 500 steps) is compared against the Townsend integrator (blue circles, $\Delta t = t_{cool}$) and the N-R solver with $\Delta_{var} = 4$ (red squares) and 20 (green triangles). The orange dashed line marks T_{eq}.

neighbour on the finer level falls below it, solely due to the time-step difference.

Undercooling bias. As shown in Fig. 4, the N-R solver with $\Delta_{var} = 4$ systematically *undercools* in the ISM regime — the computed temperature remains above the RK4 reference trajectory. This is a consequence of the semi-implicit (backward Euler) nature of the scheme: the implicit linearisation evaluates the cooling rate at the *updated* temperature, which is lower and closer to equilibrium, where $|\Lambda_{net}|$ is smaller. The result is a systematic underestimate of the integrated cooling. The bias increases with the time-step size, creating a correlation between AMR level and temperature error.

Level-dependent temperature gradient. Consider a uniform-density region being photoionisation-heated. With exact cooling, all cells converge to the same T_{eq} regardless of their AMR level. With the N-R solver, cells on coarser levels (larger Δt) systematically undercool more than cells on finer levels, creating an artificial temperature gradient across the level boundary. This gradient can trigger the Jeans criterion where no physical density enhancement exists, initiating a spurious refinement cascade.

Cumulative error. Because T_2 enters the next hydrodynamic time-step as the updated thermal energy, any bias at step n feeds back into the pressure and hence the Jeans length at step n+1. Over many time-steps, the level-dependent temperature bias accumulates, creating persistent artefacts in the temperature field that drive sustained spurious refinement.

Time-step coupling. The maximum sub-step size within `solve_cooling` is determined by the cooling time $t_{cool} = (3/2)\,nk_BT/|\Lambda_{net}|$. When Λ_{net} is large, the solver takes many small sub-steps and is accurate. But near equilibrium, $|\Lambda_{net}|$ is small, t_{cool} is large, and the solver takes fewer, larger steps — precisely where accuracy matters most for determining the equilibrium temperature.

4.1.3 Limitations of the 2D Table

Beyond the numerical integration, the cooling *rates* themselves have systematic biases:

(i) **No metallicity axis.** The RAMSES table is 2D: (n_H, T_2). Metal cooling is stored as a separate 1D function $\Lambda_Z(T)$ at solar metallicity, then scaled as $Z \cdot \Lambda_Z$. This linear scaling is correct for CIE but neglects the density-dependent ionisation corrections that become important

at $n_{\rm H} > 1\,{\rm cm}^{-3}$, where photo-ionisation by the UV background cannot penetrate and the gas is self-shielded.

(ii) **CIE assumption.** The mean molecular weight μ is computed assuming collisional ionisation equilibrium at each $(n_{\rm H}, T)$ point. This neglects time-dependent ionisation (important during rapid cooling) and photo-ionisation at low temperatures (important in the diffuse IGM at $T < 10^4$ K), where the UV background maintains a residual ionisation fraction that CIE would predict to be negligible.

(iii) **UV background model.** The default Courty model is a polynomial fit calibrated against Cloudy 07. More recent UV background models (e.g. Haardt & Madau 2012; Faucher-Giguère 2020) differ significantly at $z > 3$, particularly in the He II reionisation epoch ($z \sim 3$–4) and at the earliest epochs where the UV intensity rises steeply.

(iv) **Temperature variable $T_2 = T/\mu$.** The solver works with T_2 rather than T. Since μ changes across the ionisation transition ($\mu \approx 1.22$ for neutral gas, $\mu \approx 0.59$ for fully ionised), a small change in T_2 maps to a factor-of-two change in physical temperature T across the transition. This amplifies interpolation errors near $T \sim 10^4$ K.

4.2 Comparison on the Density–Temperature Plane

To quantify the systematic differences, we compare the RAMSES net cooling rate against the Grackle equilibrium cooling library (Smith et al. 2017), which uses the Haardt & Madau (2012) UV background and includes density-dependent metal cooling with full non-LTE ionisation corrections. We compute $\Lambda_{\rm net}(n_{\rm H}, T)$ on a grid of 60×256 points spanning $n_{\rm H} = 10^{-8}$–$10^4\,{\rm cm}^{-3}$ and $T = 10$–10^9 K at $z = 5.1$ (the redshift of the HR5 test restart).

Low metallicity ($Z = 10^{-3}\,Z_\odot$). Fig. 5 compares the two models at the simulation average metallicity (`z_ave` $= 10^{-3}$). At this low metallicity, the net rate is dominated by H/He processes and UV photo-heating.

The lower-left panel shows the log ratio of the cooling rates, $\log_{10}(|\Lambda_{\rm RAMSES}|/|\Lambda_{\rm Grackle}|)$. Contour lines mark the thermal equilibrium ($\Lambda_{\rm net} = 0$) for each model: the yellow solid line for RAMSES and the green dashed line for Grackle. Three features are apparent:

(i) The *equilibrium temperature* differs by up to $\Delta \log T \sim 0.3$–0.5 at low densities ($n_{\rm H} \lesssim 10^{-2}\,{\rm cm}^{-3}$), reflecting the different UV backgrounds. Since $\lambda_{\rm J} \propto T^{1/2}$, this translates to a $\sim$ 40–80 per cent difference in the Jeans length at the same density.

(ii) At intermediate temperatures (10^4–10^5 K), the RAMSES rate is systematically *stronger* (redder in the ratio panel) by up to an order of magnitude. This arises because the CIE assumption overpredicts the neutral hydrogen fraction at these temperatures for low-density gas where the UV background maintains significant photo-ionisation.

(iii) At $T > 10^7$ K, both models converge to the bremsstrahlung limit $\Lambda \propto T^{1/2}\, n_{\rm H}^2$ and the ratio approaches unity, as expected.

The lower-right panel shows the mean molecular weight difference $\Delta\mu = \mu_{\rm RAMSES} - \mu_{\rm Grackle}$. The RAMSES CIE μ is systematically higher than the Grackle non-LTE value at $T > 10^5$ K. At intermediate temperatures ($T \sim 10^4$–10^5 K), the Grackle μ shows a strong *density dependence* due to the UV background maintaining partial ionisation at low densities — a feature entirely absent from the RAMSES model, where $\mu(T_2)$ is independent of density. This density-dependent μ has direct implications for the $T_2 \to T$ conversion and hence for the Jeans length evaluation.

Solar metallicity ($Z \approx Z_\odot$). Fig. 6 shows the same comparison at near-solar metallicity, where metal line cooling dominates the temperature range $T \sim 10^4$–$10^{5.5}$ K. The differences between the two models are significantly amplified:

- The Grackle table includes density-dependent ionisation corrections for metal species, leading to *up to* $\sim 10\times$ *stronger cooling* in the $T \sim 10^{4.5}$–10^5 K range compared to the CC07 CIE table. This is precisely the temperature range where infalling gas cools through the virial temperature of 10^{11}–$10^{12}\,{\rm M}_\odot$ haloes.
- The equilibrium temperature contours diverge more strongly, with the RAMSES equilibrium systematically higher at $n_{\rm H} < 10^{-1}\,{\rm cm}^{-3}$. This implies that the standard RAMSES solver overestimates the temperature of the warm-hot intergalactic medium (WHIM), reducing the Jeans-based refinement trigger and potentially missing structure in filaments.
- The UV suppression factor for metal cooling differs: the Courty model applies a redshift-dependent attenuation $f_{\rm UV}(n_{\rm H}, z)$ that damps metal cooling more strongly at low densities than the Grackle self-shielding prescription.

4.3 Implications for the Jeans Criterion

The combined effect of the cooling rate differences (Figs 5–6), the integration scheme, and the redshift evolution of the UV background impacts the Jeans refinement criterion through four channels:

1. Equilibrium temperature. The Jeans length at thermal equilibrium is $\lambda_{\rm J}^{\rm eq} \propto (T_{\rm eq}/\rho)^{1/2}$. Since $T_{\rm eq}^{\rm RAMSES} \neq T_{\rm eq}^{\rm Grackle}$ (bottom-left panels), the two solvers will flag different cells. At low densities, the RAMSES equilibrium is systematically *higher* ($T_{\rm eq} \sim 2$–$3\times$ warmer), yielding a larger Jeans length and less refinement. This means the standard solver may *underestimate* the refinement needed in filaments and the circumgalactic medium.

2. Cooling overshoot near $T_{\rm eq}$. The N-R solver's tendency to overshoot $T_{\rm eq}$ (Section 4.1.2) creates transient temperature errors that affect the *instantaneous* Jeans length even if the time-averaged temperature is correct. A cell that overshoots to $T < T_{\rm eq}$ will have a transiently small $\lambda_{\rm J}$, triggering refinement that may not be needed. Conversely, a cell that oscillates above $T_{\rm eq}$ will have a transiently large $\lambda_{\rm J}$, suppressing refinement that *is* needed. The exact integrator eliminates this noise entirely.

3. Mean molecular weight. The $T_2 = T/\mu$ representation amplifies errors in μ. With the RAMSES CIE model, μ is a function of T_2 alone (density-independent). The Grackle non-LTE μ depends on both $n_{\rm H}$ and T, with $|\Delta\mu| \sim 0.3$–0.6 in the transition zone ($T \sim 10^4$–10^5 K; bottom-right panels). Since $T = T_2\,\mu$, a 50 per cent error in μ produces a 50 per cent error in T and hence a $\sim$ 22 per cent error in $\lambda_{\rm J}$, comparable to the factor-of-two ratio that separates adjacent refinement levels.

4. Redshift evolution of $T_{\rm eq}$. The thermal equilibrium temperature evolves with redshift as the UV background strengthens (towards $z \sim 2$) and then weakens (towards $z = 0$). A fixed-z Grackle table ignores this evolution, producing a systematic offset in $T_{\rm eq}$ that grows with $|z - z_{\rm table}|$. At the epoch of maximum UV background ($z \sim 2$–3), this offset can reach factors of 2–3 in $T_{\rm eq}$ if the table was computed at $z = 5$, corresponding to a $\sim$ 40–70 per cent error in $\lambda_{\rm J}$. The multi-z interpolation (Section 3.2) eliminates this systematic bias, ensuring that $T_{\rm eq}$ tracks the correct UV background at every epoch.

Table 3 summarises the systematic differences and their impact on the Jeans length.

5 THE STARBURST PROBLEM

The holdback mechanism (Section 2) blocks *all* refinement criteria at levels $\ell \geq \ell_{\rm hb}$, including the Jeans criterion. This creates two related problems:

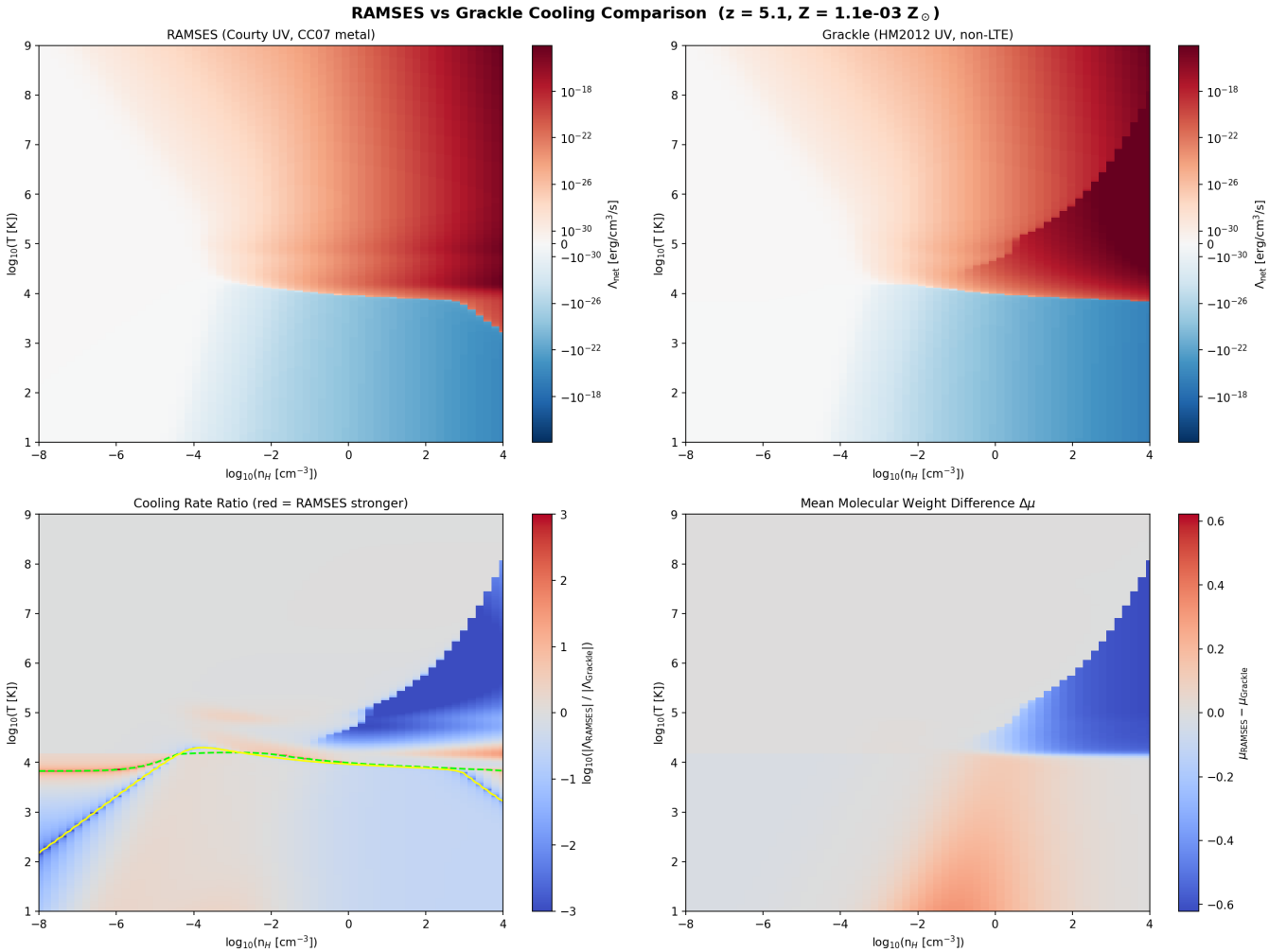

Figure 5. Comparison of the RAMSES and Grackle net cooling rates at $z = 5.1$ and $Z = 1.1 \times 10^{-3}\, Z_\odot$ (the simulation mean metallicity). *Top left:* RAMSES net cooling rate (Courty UV, Cloudy 07 CIE metals). *Top right:* Grackle net cooling rate (HM2012 UV, non-LTE equilibrium chemistry). Red shading indicates net cooling (energy loss); blue indicates net heating (energy gain). *Bottom left:* Log ratio of the absolute cooling rates, $\log_{10}(|\Lambda_{\rm RAMSES}|/|\Lambda_{\rm Grackle}|)$; yellow solid curve = RAMSES thermal equilibrium ($\Lambda = 0$), green dashed = Grackle equilibrium. Note the divergence of the equilibrium contours at $n_{\rm H} < 1\,{\rm cm}^{-3}$. *Bottom right:* Mean molecular weight difference $\mu_{\rm RAMSES} - \mu_{\rm Grackle}$, showing the density-dependent μ from Grackle's non-LTE treatment.

Table 3. Systematic differences between the RAMSES and Grackle/Townsend cooling implementations and their impact on the Jeans length.

Component	RAMSES	Grackle/Townsend	$\lambda_{\rm J}$ impact
Integration	N-R implicit (sub-stepping)	Exact Y-function (single step)	Removes overshoot noise
UV background	Courty (Cloudy 07)	Haardt & Madau (2012) (Cloudy 13)	$T_{\rm eq}$ shift $\rightarrow$ 40–80 per cent
Metal cooling	CC07 CIE, linear Z scaling	3D non-LTE, density-dep. Z	Up to 10× at $10^{4.5}$ K
μ model	CIE, density-independent	Non-LTE, $\mu(n_{\rm H}, Z, T)$	$\lvert\Delta\mu\rvert \sim 0.3$–$0.6$
Table dimension	2D: $(n_{\rm H}, T_2)$	3D: $(n_{\rm H}, Z, T)$	Metallicity resolution
Redshift	Rebuilt per step (`set_table`)	Multi-z interpolation (10 snapshots)	Tracks $T_{\rm eq}(z)$ correctly

Artificial fragmentation. The Jeans length criterion $\mathcal{J}$ differs fundamentally from the other refinement triggers: it is a *numerical stability* requirement (Truelove et al. 1997), not merely a resolution preference. Violating the Jeans condition leads to artificial fragmentation — a qualitatively wrong result — regardless of epoch. Yet in the standard RAMSES logic (equation 3), the cosmological holdback $\mathcal{H}$ blocks $\mathcal{J}$ uniformly with all other criteria. At high redshift, dense self-gravitating regions may therefore be held at insufficient resolution, producing spurious fragmentation that propagates into lower-redshift structure.

Abrupt SFR transition. When the holdback releases level ℓ (i.e. $\mathcal{H}(\ell)$ transitions from true to false), all accumulated criteria fire simultaneously. Many cells that have been gravitationally collapsing during the holdback period now suddenly satisfy the refinement condition, leading to a burst of new cells and an abrupt spike in the star formation rate.

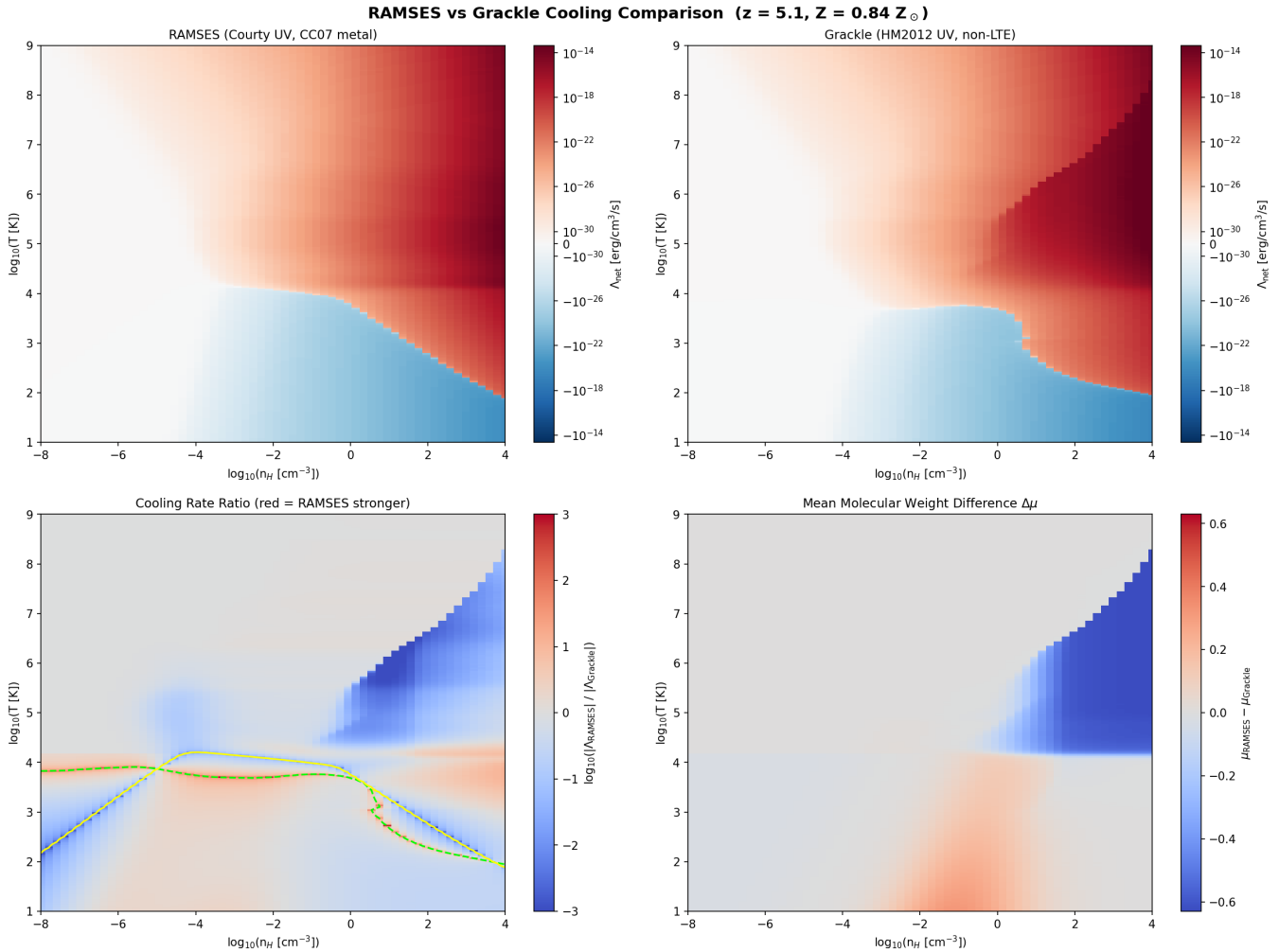

Figure 6. Same as Fig. 5 but at solar metallicity ($Z = 0.84\,Z_\odot$). Metal cooling dominates the net rate at $T \sim 10^4$–$10^{5.5}$ K, amplifying the model differences. Note the larger separation of the equilibrium contours (bottom-left) and the stronger $\Delta\mu$ pattern (bottom-right) compared to the low-metallicity case.

6 JEANS BYPASS: DESIGN AND IMPLEMENTATION

6.1 Modified Refinement Logic

The key observation is that the Jeans length criterion is *cell-local*: it depends only on the density and temperature of the cell itself (`uold`), with no neighbour information required. This allows a lightweight evaluation even when the full refinement machinery (which requires neighbour gathering for $\mathcal{D}, \mathcal{P}, \mathcal{U}$) is skipped by the holdback.

With `jeans_bypass_holdback=.true.`, the refinement logic becomes

$$\mathcal{R}' = \Big[\neg\mathcal{H} \wedge \mathcal{G} \wedge \big(\mathcal{M}\vee\mathcal{S}\vee\mathcal{D}\vee\mathcal{P}\vee\mathcal{U}\vee\mathcal{J}\big)\Big] \vee \Big[\mathcal{H} \wedge \mathcal{J} \wedge (\ell < \ell_{\rm hb}+n_{\rm bypass})\Big]; \tag{38}$$

where $n_{\rm bypass} \equiv$ `n_jeans_bypass_levels` (default 1). The geometry mask $\mathcal{G}$ still applies through the non-holdback branch, preserving zoom-in spatial restrictions. Only the temporal holdback $\mathcal{H}$ is bypassed for $\mathcal{J}$, and only within a limited number of levels.

This modification is analogous to the existing treatment of sink particles: $\mathcal{S}$ is already evaluated inside the holdback-protected block, but unlike $\mathcal{J}$ it only triggers in cells that already host a sink. Extending the same exemption to $\mathcal{J}$ ensures numerical integrity for gas that is *about to* form a sink.

6.2 Level Limitation: Preventing Cascades

Without a level limit ($n_{\rm bypass} \to \infty$), the Jeans bypass creates a catastrophic cascade. When a cell at the holdback level $\ell_{\rm hb}$ is flagged by the Jeans criterion, child cells are created at level $\ell_{\rm hb} + 1$. Since level $\ell_{\rm hb} + 1$ is also in the holdback zone, the bypass applies again, flagging cells at $\ell_{\rm hb} + 2$, and so on until $\ell_{\rm max}$. This rapidly exhausts available memory.

The cascade occurs because the Jeans length in radiatively cooling gas (below the star-formation density threshold) shrinks without bound as the temperature drops. There is no physical mechanism to halt the Jeans criterion from triggering at every successive level — only the polytropic temperature floor (Section 7) or a hard level limit can stop it.

Table 4 shows the progression observed in a test run without level limit.

The memory footprint grew from 81 per cent to 88.5 per cent in a single coarse step before crashing. The fine-step time-step also decreased by 8× due to subcycling at levels 15–17.

Table 4. Cascade refinement without level limit ($a = 0.164$, $\ell_{\rm hb} = 14$, $\ell_{\rm max} = 17$). All levels ≥ 14 are in the holdback zone.

Level	$\Delta x/\Delta x_{\rm min}$	Holdback	Bypass	$\mathcal{J}$ fires	Result
≤ 12	≥ 32	No	—	Yes (normal)	Normal
13	16	No	—	Yes (normal)	Normal
14	8	Yes	Active	Yes	→ lv. 15 created
15	4	Yes	Active	Yes	→ lv. 16 created
16	2	Yes	Active	Yes	→ lv. 17 created
17	1	$\ell_{\rm max}$	—	—	OOM crash

Table 5. Polytropic temperature floor and Jeans length behaviour for different values of $g_\star$, evaluated at the star-formation threshold $n = 55\,n_\star$.

$g_\star$	$T_{\rm floor}$ (K)	$\lambda_{\rm J}$ scaling	Cascade safe?
5/3	~10 900	$\rho^{-1/6}$ (decreasing)	No
2	~41 300	const	Yes
7/3	~157 000	$\rho^{+1/6}$ (increasing)	Yes

With $n_{\rm bypass} = 1$, only level $\ell_{\rm hb} = 14$ is bypassed, creating cells at level 15 but no further. This provides a controlled pre-refinement before the holdback releases.

7 ROLE OF THE POLYTROPIC TEMPERATURE FLOOR

7.1 Jeans Length in the Polytropic Regime

In RAMSES, the star-formation module enforces a polytropic equation of state above the density threshold $n_\star$:

$$T(n) = T_{2,\star}\left(\frac{n}{n_\star}\right)^{g_\star - 1}, \qquad n \geq n_\star, \tag{39}$$

with default parameters $T_{2,\star} = 750\,\mathrm{K}$, $n_\star = 0.1\,\mathrm{cm}^{-3}$, $g_\star = 5/3$. This floor is applied in the cooling routine and is already reflected in `uold`, so the Jeans length computation in `jeans_length_refine()` automatically includes the polytropic stabilisation.

Substituting equation (39) into equation (8):

$$\lambda_{\rm J} \propto \sqrt{\frac{T}{\rho}} \propto \sqrt{\rho^{g_\star - 1}/\rho} = \rho^{(g_\star - 2)/2}. \tag{40}$$

The behaviour of $\lambda_{\rm J}$ in the polytropic regime depends critically on the value of $g_\star$:

- $g_\star < 2$: $\lambda_{\rm J}$ decreases with density. For the standard value $g_\star = 5/3$, $\lambda_{\rm J} \propto \rho^{-1/6}$ — the decrease is slow, but unbounded. Given enough levels, a cascade can still develop.
- $g_\star = 2$: $\lambda_{\rm J} \propto \rho^0 = \mathrm{const}$. The Jeans length is *independent of density* in the polytropic regime, so refinement converges at a fixed level regardless of how much the gas compresses. This is the *critical polytropic index* for Jeans stability.
- $g_\star > 2$: $\lambda_{\rm J}$ increases with density, providing an even stronger self-regulation.

Table 5 quantifies the temperature floor at the star-formation threshold $n = \texttt{del_star} \times n_\star = 55\,n_\star$ for different values of $g_\star$, keeping $T_{2,\star} = 750\,\mathrm{K}$.

The $g_\star = 2$ case is appealing because it eliminates the cascade problem entirely without any additional mechanisms (no holdback, no `d_jeans_thre`, no level limits). However, it comes at a significant physical cost: the temperature floor at the star-formation threshold rises from ~10 900 K to ~41 300 K — nearly a factor of 4. This dramatically increases the pressure support in star-forming gas, suppressing the peak densities and reducing the star formation rate. Since $\dot{\rho}_* \propto \rho/t_{\rm ff} \propto \rho^{3/2}$, even modest changes in the density PDF propagate into order-of-magnitude SFR differences. Adopting $g_\star = 2$ therefore requires a full recalibration of the star-formation parameters (ϵ_*, `del_star`, $T_{2,\star}$, $n_\star$) against observational constraints, which is non-trivial for production simulations that have already been calibrated with $g_\star = 5/3$.

Table 6. Two complementary mechanisms for preventing the bypass cascade.

Mechanism	Type	Behaviour
`n_jeans_bypass_levels`	Hard limit	Caps extra levels regardless of physics
`d_jeans_thre` $\approx n_\star$	Physics-based	Polytropic floor limits $\lambda_{\rm J}$ decrease

In practice, the standard choice $g_\star = 5/3$ combined with `d_jeans_thre` $\approx n_\star$ provides an effective compromise: the Jeans criterion is restricted to the polytropic regime where $\lambda_{\rm J}$ varies slowly ($\propto \rho^{-1/6}$), limiting the cascade to 1–2 extra levels, while preserving the calibrated star-formation physics.

In contrast, for gas below $n_\star$ (no polytropic floor), radiative cooling can drive T to very low values while ρ remains moderate. This yields $\lambda_{\rm J} \ll \Delta x_\ell$ at many levels, causing the cascade.

7.2 Self-regulation via `d_jeans_thre`

The density threshold parameter `d_jeans_thre` (in units of $n_{\rm H}$ [cm^{-3}]) restricts the Jeans criterion to cells above a given density:

$$\mathcal{J}(\text{cell}) = \begin{cases} N_J\,\Delta x_\ell \geq \lambda_{\rm J} & \text{if } n \geq \texttt{d_jeans_thre}, \\ \text{false} & \text{otherwise.} \end{cases} \tag{41}$$

Setting `d_jeans_thre` $\approx n_\star$ restricts the Jeans criterion to the polytropic regime. The cascade then self-limits through the following mechanism:

(i) At level $\ell_{\rm hb}$: only cells with $n \geq n_\star$ are checked. The polytropic floor guarantees $T \geq T_{2,\star}$, so $\lambda_{\rm J}$ is relatively large.

(ii) If $N_J\,\Delta x_{\ell_{\rm hb}} > \lambda_{\rm J}$, the cell is flagged and children at $\ell_{\rm hb} + 1$ are created.

(iii) At level $\ell_{\rm hb} + 1$: Δx is halved, density is approximately unchanged, $\lambda_{\rm J}$ is approximately unchanged.

(iv) $N_J\,\Delta x_{\ell_{\rm hb}+1} = N_J\,\Delta x_{\ell_{\rm hb}}/2 < \lambda_{\rm J} \Rightarrow$ Jeans criterion is *satisfied* $\Rightarrow$ no further flagging $\Rightarrow$ cascade stops naturally.

This provides a *physics-based* self-regulation, complementary to the *hard limit* imposed by `n_jeans_bypass_levels`.

When `d_jeans_thre` = 0 (default), *all* cells are checked, including low-density gas where cooling produces arbitrarily small $\lambda_{\rm J}$. This is the root cause of the unlimited cascade observed in Section 6.2.

8 IMPLEMENTATION DETAILS

8.1 Code Structure

The implementation modifies two source files (plus parameter declaration and namelist reading).

`flag_utils.kjhan.f90` — `userflag_fine()`. The holdback early-return is modified from a simple `return` to a conditional block:

```
if(prevent_refine .and. q_refine_holdback) then
   if(jeans_bypass_holdback .and. hydro &
      .and. n_jeans_bypass_levels > 0) then
      ! Find the first holdback level
```

```
      dx_min = (0.5D0**nlevelmax)*scale
      holdback_threshold = &
        (4d0**(1d0/ndim))*(dx_min/aexp)
      holdback_first = nlevelmax + 1
      do jlevel = nlevelmax_part &
         + nlevel_collapse+1, nlevelmax
         dx_test = (0.5d0**jlevel)*scale
         if(dx_test < holdback_threshold) then
            holdback_first = jlevel
            exit
         endif
      end do
      ! Only bypass within allowed range
      if(ilevel < holdback_first &
         + n_jeans_bypass_levels) then
         call jeans_only_flag(ilevel)
      endif
   endif
   return
endif
```

`hydro_flag.kjhan.f90` — `jeans_only_flag()`. A new lightweight subroutine that:

(i) Loops over active grids with OpenMP parallelism.
(ii) Computes cell indices without neighbour gathering.
(iii) Calls the existing `jeans_length_refine()` subroutine (no code duplication).
(iv) Sets `flag1` for flagged cells and updates `nflag`.

This is possible because `jeans_length_refine()` only accesses cell-local data (`uold(ind_cell,...)`), unlike the gradient-based criteria $\mathcal{D}, \mathcal{P}, \mathcal{U}$.

8.2 Safety Properties

- **Default off**: With `jeans_bypass_holdback=.false.` (default), the new code path is never entered. Bit-identical to the unmodified code.
- **No neighbour access**: `jeans_only_flag()` is safe to call at any point in the refinement cycle.
- **OMP safety**: `flag1` writes are grid-partitioned; `nflag` uses reduction.
- **Hard level limit**: `n_jeans_bypass_levels` provides a fail-safe even if `d_jeans_thre` is misconfigured.

8.3 Summary of Parameters

Table 7 lists all refinement-related namelist parameters, including the new bypass parameters.

The recommended configuration for cosmological runs with star formation is:

```
&REFINE_PARAMS
m_refine           = 9*8.,
jeans_refine       = 9*4.,
jeans_bypass_holdback  = .true.
n_jeans_bypass_levels  = 1
d_jeans_thre       = 0.1      ! = n_star
q_refine_holdback = .true.
/
```

9 TEST RESULTS

All tests use the HR5 medium-box setup: $\ell_{\rm min} = 9$, $\ell_{\rm max} = 17$, 12 MPI ranks, OMP=5, restarting from output 5 ($a = 0.164$, $z \approx 5.1$, nstep_coarse=241).

9.1 Bit-identical Verification

With `jeans_bypass_holdback=.false.` (default) and no `jeans_refine`, the new binary produces results identical to the reference run: econs=6.13×10^{-4} at step 242, econs=6.71×10^{-4} at step 243.

9.2 Unlimited Bypass

With `jeans_refine=9*4.`, `jeans_bypass_holdback=.true.`, `d_jeans_thre=0`, and no level limit: cascade refinement from level 14 through 17 within a single coarse step. Memory grew from 81 per cent to 88.5 per cent before OOM crash. Fine-step time-step decreased by 8× due to subcycling.

9.3 Limited Bypass ($n_{\rm bypass} = 1$)

With `n_jeans_bypass_levels=1`: only level 14 is bypassed, cells at level 15 may be created but no further cascade.

9.4 Future: `d_jeans_thre` $= n_\star$

A test with `d_jeans_thre=0.1` will verify the physics-based self-regulation, potentially allowing $n_{\rm bypass} > 1$ without cascade, since only gas in the polytropic regime would trigger the Jeans criterion.

10 DISCUSSION AND FUTURE WORK

10.1 Physical Motivation

The holdback mechanism was designed to prevent over-refinement from the mass criterion at early times. However, it inadvertently blocks the Jeans criterion, which has a different physical motivation: resolving the Jeans length prevents artificial fragmentation and ensures numerically converged star formation (Truelove et al. 1997; Federrath et al. 2011).

By allowing the Jeans criterion to operate ahead of the holdback release, we expect:

- Dense gas that will form stars is pre-refined before the holdback releases.
- When the holdback releases, the transition is smoother because the densest regions are already at the appropriate resolution.
- The SFR curve shows a gradual onset rather than an abrupt step.

10.2 Interplay of the Three Parameters

The bypass is controlled by three parameters that form a layered defence:

Layer 1: `jeans_bypass_holdback` (on/off). Master switch. When off (default), the system is identical to standard RAMSES.

Layer 2: `d_jeans_thre` (physics-based regulation). Restricts the Jeans check to dense gas where the polytropic floor operates. In this regime, $\lambda_{\rm J} \propto \rho^{(g_\star-2)/2}$ varies slowly (for $g_\star = 5/3$, $\propto \rho^{-1/6}$), and 1–2 extra refinement levels suffice to resolve it. As shown in Section 7, adopting $g_\star \geq 2$ would eliminate the cascade entirely, but at the cost of a significantly stiffer pressure floor that requires SFR recalibration (Table 5). Provides self-consistent, physics-driven cascade termination.

Layer 3: `n_jeans_bypass_levels` (hard safety limit). Absolute cap on the number of extra levels. Prevents OOM regardless

Table 7. AMR refinement parameters in the `&AMR_PARAMS` and `&REFINE_PARAMS` namelists. New parameters are marked with ⋆.

Parameter	Default	Role
`m_refine`	8.0	Lagrangian mass threshold (per level)
`sink_refine`	F	Force max level around sinks
`err_grad_d`	−1	Density gradient threshold
`err_grad_p`	−1	Pressure gradient threshold
`err_grad_u`	−1	Velocity gradient threshold
`jeans_refine`	−1	Jeans length in cells (per level)
`d_jeans_thre`	0	Density threshold for Jeans ($n_{\rm H}$ cm^{-3})
`r_refine`	−1	Geometry region diameter (per level)
`a_refine`	1.0	Ellipticity Y/X
`b_refine`	1.0	Ellipticity Z/X
`exp_refine`	2.0	Norm exponent
`x/y/z_refine`	0.5	Region centre
`nexpand`	1	Buffer zone width (per level)
`q_refine_holdback`	F	Cosmological holdback
⋆ `jeans_bypass_holdback`	F	Enable Jeans bypass of holdback
⋆ `n_jeans_bypass_levels`	1	Max extra levels beyond $\ell_{\rm hb}$

of physics configuration. Essential when `d_jeans_thre=0` (all cells checked).

10.3 Problems with Additional Temperature and Pressure Floors

A common approach to enforcing the Jeans criterion in cosmological simulations is to introduce an *additional* temperature or pressure floor beyond the star-formation polytropic floor (equation 39). Several major simulation projects adopt a resolution-dependent pressure floor of the form

$$P_{\rm floor} = \frac{G}{\pi\,\gamma}\,N_J^2\,\rho\,(\Delta x_\ell)^2, \tag{42}$$

which ensures $\lambda_{\rm J} \geq N_J\,\Delta x_\ell$ at all times (e.g. Schaye et al. 2015; Hopkins et al. 2018). Equivalently, this corresponds to a density-dependent temperature floor $T_{\rm floor} \propto \rho\,(\Delta x_\ell)^2$. While this *guarantees* that the Jeans criterion is never violated, it introduces several systematic problems:

(i) Artificial suppression of fragmentation. The pressure floor prevents gas from cooling below $T_{\rm floor}$, even when the physical equilibrium temperature $T_{\rm eq}$ is much lower. In the density range $n_\star < n < n_{\rm floor}$ (where $n_{\rm floor}$ is the density at which $T_{\rm floor}$ exceeds the polytropic floor), the gas is held at an artificially high temperature. This suppresses the thermal instability that drives the formation of the cold, dense phase of the ISM (Sutherland & Dopita 1993; Wiersma, Schaye & Smith 2009), potentially preventing the formation of molecular clouds and reducing the star formation rate.

(ii) Resolution-dependent thermodynamics. Since $P_{\rm floor} \propto (\Delta x_\ell)^2$, the effective equation of state of the gas depends on the grid resolution. At higher resolution ($\ell_{\rm max}$ increased), Δx decreases and the floor drops, allowing gas to cool further and fragment into smaller structures. The resulting star formation rate, stellar mass function, and galaxy morphologies are therefore resolution-dependent — not converging to a physical answer as resolution increases, but rather revealing new fragmentation at each resolution level.

(iii) Conflict with the polytropic floor. The star-formation polytropic floor (equation 39) is designed to model the unresolved physics of the multiphase ISM above $n_\star$: it represents the effective equation of state of a turbulent, star-forming medium (Schaye et al. 2015). An additional Jeans pressure floor that is active *below* $n_\star$ creates a discontinuity in the effective equation of state at $n = n_\star$ if the two floors do not match. More fundamentally, the Jeans floor operates on *numerical* grounds (preventing artificial fragmentation) while the polytropic floor operates on *physical* grounds (modelling the unresolved ISM), and conflating the two obscures the distinction between numerical and physical effects.

(iv) Masking of cooling errors. The pressure floor removes the consequences of cooling inaccuracies (Section 4.1.2) by overriding the computed temperature whenever it falls below $T_{\rm floor}$. While this prevents the most extreme manifestation of cooling artefacts (spurious fragmentation), it also prevents the diagnosis of the underlying problem. The Δt-dependent temperature biases of the N-R solver create artificial temperature discontinuities at AMR level boundaries that the floor cannot address, since the erroneous temperatures typically remain above $T_{\rm floor}$. Over many time-steps, these hidden biases accumulate and affect the refinement pattern and the large-scale thermodynamic state.

(v) Incompatibility with turbulent Jeans analysis. If the Jeans criterion is to be extended to include turbulent pressure support (equation 2), the floor must also be modified to account for $\sigma_{\rm turb}$. But the turbulent velocity dispersion varies in space and time, making a simple analytic floor impossible. The approach advocated in this paper — *resolving* the Jeans length through AMR rather than *imposing* it through a floor — is naturally compatible with a turbulence-dependent criterion, since the refinement condition (equation 8) can be straightforwardly generalised to use $c_{\rm eff}^2 = c_s^2 + \sigma_{\rm turb}^2$ without modifying any thermodynamic floor.

In summary, while a pressure floor provides a brute-force guarantee that the Jeans criterion is satisfied, it does so at the cost of altering the thermodynamic state of the gas. Our approach — combining accurate cooling (Sections 3–4) with Jeans-aware refinement (Section 6) — achieves the same numerical stability without modifying the gas physics.

10.4 The eEOS Polytropic Energy Floor

While the pressure floor approach of equation (42) modifies the thermodynamic state in a resolution-dependent way, we find that a *minimal* energy floor based on the existing polytropic equation of state is both necessary and sufficient to prevent catastrophic simulation failure when the cosmological holdback is disabled. We present here the motivation, implementation, and numerical verification of this energy floor.

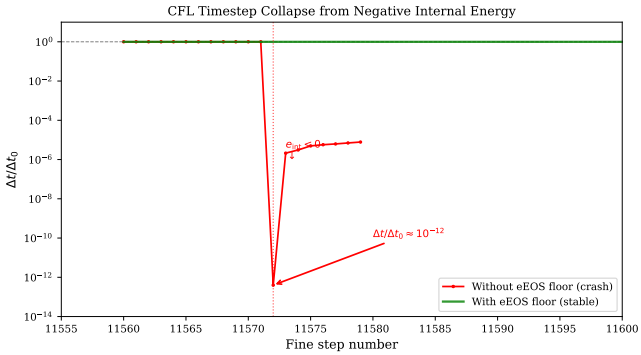

Figure 7. CFL time-step collapse caused by negative internal energy. Red: without the eEOS floor, the time-step drops by ~ 12 orders of magnitude at fine step 11572 (coarse step 408), effectively halting the simulation. Green: with the eEOS polytropic floor, the time-step remains stable throughout. The crash occurs at $z \approx 3.7$ when the cosmological holdback is disabled.

10.4.1 The Negative Internal Energy Problem

When holdback is disabled (`q_refine_holdback = .false.`), the AMR hierarchy resolves structure at higher levels earlier than in the standard configuration. This creates conditions where the hydro solver can produce negative internal energy through the following mechanism. In a total-energy formulation, the internal energy is computed as a residual:

$$e_{\rm int} = E_{\rm tot} - \tfrac{1}{2}\rho\,|\mathbf{v}|^2, \tag{43}$$

where $E_{\rm tot}$ is the conserved total energy density. When the kinetic energy dominates ($\frac{1}{2}\rho|\mathbf{v}|^2 \gg e_{\rm int}$), small truncation errors in the total energy or momentum can produce a negative residual. This situation arises naturally in high-Mach-number flows driven by supernova and AGN feedback, which inject large kinetic energies into ambient gas. Without the holdback, such flows are better resolved at fine levels, leading to sharper velocity gradients and a higher probability of the kinetic energy exceeding the total energy after a Riemann flux update.

Once $e_{\rm int} < 0$ in even a single cell, the consequences are catastrophic. The pressure $P = (\gamma - 1)\,e_{\rm int}$ becomes negative, and the sound speed

$$c_s = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma(\gamma-1)\,e_{\rm int}}{\rho}} \tag{44}$$

becomes imaginary. The CFL time-step

$$\Delta t = C_{\rm CFL}\,\frac{\Delta x}{|v| + c_s} \tag{45}$$

collapses to effectively zero (in practice, $\Delta t/\Delta t_0 \sim 10^{-12}$), halting the simulation.

Figure **??** illustrates this phenomenon from our diagnostic test. At fine step 11572 of coarse step 408 ($z \approx 3.7$, $a = 0.212$), the global internal energy drops to $e_{\rm int} = -2.70 \times 10^{-3}$ (code units). The CFL time-step immediately collapses from $\Delta t = 1.76 \times 10^{-4}$ to $\Delta t = 7.0 \times 10^{-17}$, a factor of $\sim 10^{12}$. After this point, the simulation is effectively frozen, as each subsequent time-step advances by $\lesssim 10^{-9}$ code time units.

10.4.2 Design of the Energy Floor

The energy floor enforces a minimum internal energy density based on the existing star-formation polytropic equation of state:

$$e_{\rm int} \geq e_{\rm floor} \equiv \frac{T_{2,\star}}{\mu\,(\gamma-1)}\,\frac{k_{\rm B}}{m_{\rm H}}\left(\frac{\rho}{\rho_\star}\right)^{g_\star - 1}\rho, \tag{46}$$

or equivalently in code units:

$$e_{\rm int} \geq C_{\rm eEOS}\,\rho^{g_\star}, \tag{47}$$

where $C_{\rm eEOS}$ is a single coefficient computed at initialisation from the simulation parameters ($T_{2,\star}$, $n_\star$, $g_\star$, and the unit conversions):

$$C_{\rm eEOS} = \frac{T_{2,\star}}{(\gamma-1)\,{\rm scale}_{T_2}\,{\rm scale}_{n_{\rm H}}^{g_\star - 1}}. \tag{48}$$

For the standard parameters ($T_{2,\star} = 750\,{\rm K}$, $n_\star = 0.1\,{\rm cm}^{-3}$, $g_\star = 5/3$), this gives $C_{\rm eEOS} = 4.97 \times 10^{-11}$ in code units.

Crucially, this floor is *not* an additional pressure floor in the sense of equation (42). It is the natural extension of the existing polytropic floor to all densities: above $n_\star$, it is already enforced by the cooling routine; below $n_\star$, it provides a minimal energy guarantee that prevents the total-energy formulation from producing negative pressures. Because the polytropic scaling $e_{\rm floor} \propto \rho^{5/3}$ implies $T_{\rm floor} \propto \rho^{2/3}$, the floor temperature drops rapidly at low densities (e.g. $T_{\rm floor} \sim 10\,{\rm K}$ at $n \sim 10^{-4}\,{\rm cm}^{-3}$), far below the photoionisation equilibrium temperature ($T_{\rm eq} \sim 10^4\,{\rm K}$). The floor therefore has no effect on the thermodynamics of diffuse gas in normal conditions — it only activates when a catastrophic error would otherwise crash the simulation.

Figure **??** shows the temperature–density relation and the corresponding Jeans length for three cases: cooling equilibrium only, standard polytropic floor (active only above $n_\star$), and the eEOS floor (active at all densities). In the diffuse regime ($n \ll n_\star$), the eEOS floor lies far below $T_{\rm eq}$ and has no effect on either the thermodynamics or the Jeans length. Above $n_\star$, the eEOS floor coincides with the standard polytropic floor. The protection is only relevant in the narrow regime where numerical errors would otherwise drive $e_{\rm int}$ below zero.

10.4.3 Implementation

The energy floor is applied at three points in the time integration cycle, corresponding to the three operations that can modify the internal energy:

(i) **Primitive variable reconstruction** (`ctoprim` in `umuscl.f90`): Before the Riemann solver computes fluxes, the left and right states are reconstructed from conservative variables. The internal energy of each state is clamped to $\max(e_{\rm int}, C_{\rm eEOS}\,\rho^{g_\star})$. This prevents the Riemann solver from encountering negative pressures in its input states.

(ii) **Post-flux update** (`set_uold` in `godunov_fine.f90`): After the conservative update $\mathbf{U}^{n+1} = \mathbf{U}^n - \Delta t\,\nabla\cdot\mathbf{F}$, the internal energy is recomputed from the residual (equation **??**) and clamped. If the floor is activated, only the energy is modified; the momentum is left unchanged to preserve the flow field.

(iii) **Post-feedback enforcement** (`enforce_eeos_after_sink` in `amr_step.f90`): After AGN feedback deposits energy (which can produce extreme velocities via kinetic-only injection when $T > T_{2,\rm maxAGN}$), a two-stage protection is applied. First, the eEOS floor is enforced; if the floor is activated, the excess kinetic energy that would be needed to restore the floor is removed by scaling down

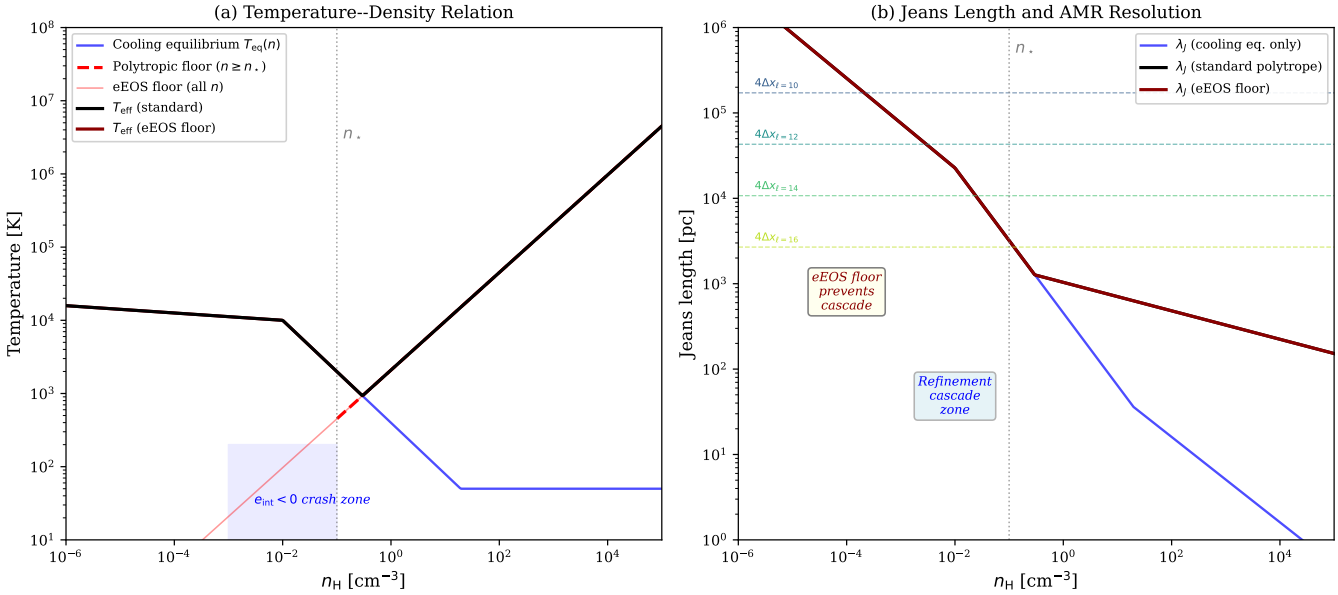

Figure 8. (a) Temperature–density relation showing the cooling equilibrium $T_{eq}(n)$, the standard polytropic floor (active only for $n \geq n_\star$), and the eEOS floor (active at all densities). At low densities, the eEOS floor lies far below the cooling equilibrium and does not affect the gas thermodynamics. (b) Corresponding Jeans length λ_J for each temperature model. Horizontal dashed lines show $4\Delta x_\ell$ for selected AMR levels. The eEOS floor raises the minimum Jeans length at intermediate densities, preventing the refinement cascade that would otherwise develop below $n_\star$.

Table 8. Comparison of the holdback-off simulation with and without the eEOS polytropic energy floor. Both runs restart from the same snapshot at $z \approx 4$ (step 377).

Quantity	Without eEOS	With eEOS
Δt range	7.0×10^{-17}–1.9×10^{-4}	1.47×10^{-4}–1.93×10^{-4}
Δt ratio (max/min)	2.8×10^{12}	1.31
$e_{\rm int,\,min}$	-2.70×10^{-3}	$+1.63 \times 10^{-9}$
Steps completed	408 (crash)	416+ (stable)
DIAG triggers ($e_{\rm int} < 0$)	—	0

the momentum (preserving total energy). Second, a Mach number cap ($\mathcal{M} \leq 100$) converts excess kinetic energy above $100\,c_s$ into thermal energy, modelling instantaneous shock thermalisation.

10.4.4 Diagnostic Verification

To verify the effectiveness of the energy floor, we performed a controlled diagnostic test. Starting from a common restart point at $z \approx 4$ (coarse step 377), we inserted minimum internal energy checks after each major physics operation: SNII feedback, AGN feedback, sink creation, the Godunov hydro solver, cooling, and Bondi accretion. Table **??** summarises the results.

Over 39 coarse steps (377–416), the eEOS-protected run maintained $e_{\rm int,\,min} > 0$ at every diagnostic checkpoint, with all 117 coarse-level checks returning values in the range 1.2–1.7×10^{-9}. No per-level (fine-level) diagnostic was triggered, confirming that the floor prevented negative internal energy across all AMR levels and all physics modules. The CFL time-step remained in the narrow range 1.47–1.93×10^{-4}, compared to the holdback-on value of 3.5–5.1×10^{-4}. The factor-of-2.6 reduction in Δt is a natural consequence of the finer spatial resolution achieved without the holdback (smaller cells $\rightarrow$ smaller CFL Δt), not a sign of numerical instability.

10.5 Turbulence and the Effective Jeans Length

The analysis presented in this paper treats the Jeans length as a purely thermal quantity (equation 8). In reality, turbulent motions in the ISM provide additional pressure support that modifies the effective stability criterion. Following Chandrasekhar (1951) and Bonazzola et al. (1987), the turbulent Jeans length is

$$\lambda_J^{\rm turb} = \sqrt{\frac{\pi\,(c_s^2 + \sigma_{\rm turb}^2)}{G\,\rho}}, \tag{49}$$

where $\sigma_{\rm turb}$ is the turbulent velocity dispersion on the grid scale. In the limit $\sigma_{\rm turb} \gg c_s$, the turbulent Jeans length can exceed the thermal Jeans length by an order of magnitude, fundamentally changing the refinement pattern.

Subgrid-scale turbulence models (Schmidt, Niemeyer & Hillebrandt 2006; Schmidt & Federrath 2011) evolve a turbulent energy density $e_{\rm turb}$ alongside the hydrodynamic variables, providing a local estimate of $\sigma_{\rm turb} = (2e_{\rm turb}/\rho)^{1/2}$. Rennehan (2021) applied this approach to cosmological simulations, showing that SGS turbulence can suppress artificial fragmentation in the circumgalactic medium. Semenov, Kravtsov & Gnedin (2016) demonstrated that a turbulence-regulated star formation prescription produces more realistic star formation histories than a simple density-threshold model.

In galactic discs, turbulence is driven by a combination of supernova feedback, gravitational instabilities, magnetorotational instability, and large-scale shear (Robertson & Kravtsov 2008; Kim & Ostriker 2017; Hopkins et al. 2018). The relative importance of these drivers varies with environment: supernova feedback dominates in star-forming regions, while gravitational infall and accretion shocks drive turbulence in the CGM and IGM (Iapichino et al. 2011).

All three requirements are now implemented in the SGS turbulence model described in Section 2.3: the SGS energy provides a local estimator of $\sigma_{\rm turb}$ (equation 14), the Jeans criterion uses $c_{\rm eff}^2 = c_s^2 + \sigma_{\rm turb}^2$ (equation 28), and the equilibrium analysis (Section 2.3.2) shows

Table 9. Comparison of cascade prevention by turbulence and holdback. The turbulent Mach numbers are representative values for each environment.

Environment	$\mathcal{M}_{\rm turb}$	$\Delta\ell_{\rm turb}$	$\Delta\ell_{\rm hb}(z{=}5)$	Sufficient?
IGM ($z \sim 5$)	0.1–0.5	0.0–0.2	3.8	No
Filaments	1–3	0.5–1.7	3.8	No
Halo virial region	1–3	0.5–1.7	3.8	No
Cold streams	3–10	1.7–3.3	3.8	Marginal
ISM (cold, dense)	5–10	2.3–3.3	—[a]	N/A

[a] Polytropic floor already prevents cascade in this regime.

that the cascade terminates naturally at $N_{J,\rm eff} \approx 3.5$ cells per Jeans length.

Table 10. Comparison of one coarse time-step at $z \approx 9$ with and without the holdback mechanism, both using the exact cooling integrator with multi-z table interpolation and $N_J = 4$.

Quantity	Holdback ON	Holdback OFF
Active levels	12–14	12–17
Wall time (s)	125	418
Grid memory fraction	18.3%	24.9%
Stars formed	245	277
Sinks formed	25	25
Fine time-step Δt	3.9×10^{-3}	4.0×10^{-4}
Star formation levels	12–14	12–17

10.6 Can Turbulence Replace the Holdback?

A natural question is whether turbulent pressure support alone can prevent the refinement cascades that the holdback mechanism was designed to suppress, thereby rendering the holdback unnecessary. We present both a quantitative scaling analysis and a direct numerical test.

Scaling analysis. Including turbulence, the Jeans length increases by a factor $\lambda_{\rm J}^{\rm turb}/\lambda_{\rm J} = (1 + \mathcal{M}_{\rm turb}^2)^{1/2}$, where $\mathcal{M}_{\rm turb} = \sigma_{\rm turb}/c_s$ is the turbulent Mach number. In terms of refinement levels, turbulence prevents the cascade from extending by

$$\Delta\ell_{\rm turb} = \tfrac{1}{2}\log_2(1 + \mathcal{M}_{\rm turb}^2) \tag{50}$$

levels. Table 8 compares this with the holdback, which prevents approximately $\Delta\ell_{\rm hb} \approx \log_2(4^{1/3}/a_{\rm exp})$ levels at a given expansion factor.

At $z = 5$ ($a_{\rm exp} = 1/6$), the holdback suppresses ~3.8 levels of refinement. Turbulence can match this only for $\mathcal{M}_{\rm turb} \gtrsim 10$, which occurs exclusively in the cold, dense ISM — where the polytropic floor already prevents the cascade. In the diffuse IGM and filaments where the holdback is most critical, typical Mach numbers are $\mathcal{M}_{\rm turb} \sim 0.1$–3, providing at most 1–2 levels of protection. Furthermore, the resolved velocity dispersion scales as $\sigma_{\rm turb}(\ell) \propto \Delta x^{1/3}$ for Kolmogorov turbulence, so turbulent support weakens at finer levels, exacerbating the mismatch.

We therefore conclude that *resolved turbulence alone cannot replace the holdback*; the resolved velocity dispersion is too weak in the IGM where the holdback is most needed. The refinement-only SGS turbulence model (Section 2.3) provides an alternative approach: it evolves the unresolved turbulent energy dynamically and uses it in the Jeans criterion, providing a scale-dependent pressure floor that adapts to the local flow conditions. Crucially, because the SGS energy does not enter the Riemann solver (in the default `sgs_hydro=.false.` mode), it imposes no CFL penalty on the hydrodynamic time-step. The SGS model and the holdback are therefore complementary: the SGS provides physically motivated refinement control, while the holdback remains available as a safety net for regimes where the SGS model has not yet reached equilibrium.

10.7 Holdback as a Workaround for Cooling Errors

The analysis above assumes that all Jeans-triggered refinements at high redshift are physically motivated. However, as demonstrated in Section 4.1.2, the standard Newton–Raphson cooling solver produces *time-step-dependent* temperatures: cells at different AMR levels receive different Δt and therefore converge to different temperatures even under identical physical conditions. These artificial temperature discontinuities at level boundaries create spurious Jeans-length gradients that trigger refinement where no physical density enhancement exists. Similarly, a single-redshift cooling table applied at the wrong epoch can shift $T_{\rm eq}$ by factors of ~2 (Section 3.2). Together, these numerical artefacts generate *spurious* Jeans triggers that dominate the refinement budget at high z.

With a correct cooling implementation — the Townsend exact integrator providing Δt-independent temperatures, and multi-redshift table interpolation providing the appropriate UV background at each epoch — all cells at the same physical conditions converge to the same temperature regardless of their AMR level. The artificial temperature discontinuities at level boundaries vanish, eliminating spurious Jeans triggers. Most diffuse gas at high redshift remains in photoionisation equilibrium at $T \approx 10^4$ K. The Jeans length in this gas is large ($\lambda_{\rm J} \sim 80$ kpc at $n_{\rm H} = 0.01\,{\rm cm}^{-3}$), well above the refinement threshold, and the Jeans criterion does not trigger. Only genuinely self-shielded, collapsing gas triggers refinement — which is the physically correct behaviour.

Numerical test. We test this hypothesis directly by running two cosmological simulations from $z = 50$ to $z \approx 9$, restarting from identical initial conditions with the Jeans criterion active ($N_J = 4$ at all levels):

(i) **Holdback ON + exact cooling**: standard configuration with the Townsend integrator and multi-z tables;

(ii) **Holdback OFF + exact cooling**: identical, but with `q_refine_holdback = .false.`

Both runs use a 512^3 base grid ($\ell_{\rm min} = 9$, $\ell_{\rm max} = 17$) with 12 MPI ranks. Table 9 summarises the results after one coarse time-step at $z \approx 9$.

The key result is that the simulation *without holdback did not undergo a refinement cascade*. Grid memory usage increased by only 36 per cent (24.9% vs. 18.3%), and the total number of sinks (the most massive collapsed objects) remained identical at 25. The 3.3× increase in wall time is entirely attributable to the CFL constraint at level 17 (the fine time-step is ~10× smaller), not to an explosion in the number of cells. Stars form at level 17 in the holdback-off run — physically appropriate resolution for the densest, self-shielded gas.

These results support the interpretation that the cosmological holdback mechanism has historically been compensating for the Δt-dependent temperature artefacts of the N-R solver (Section 4.1.2) rather than preventing physically motivated refinement. With a Δt-independent cooling implementation, the artificial temperature discontinuities at level boundaries vanish, and the holdback is no longer necessary for *stability*; its remaining role is purely one of *computational cost control*.

11 CONCLUSIONS

We have presented three improvements to the handling of the Jeans criterion in cosmological AMR simulations, implemented in the CURAMSES code:

(i) **Improved cooling module.** The Eunha cooling module replaces the standard RAMSES Newton–Raphson solver with the Townsend (2009) exact integration scheme applied to 3D Grackle equilibrium cooling tables (Smith et al. 2017). The exact integrator is genuinely time-step independent (Section 3), eliminating the artificial temperature discontinuities at AMR level boundaries that arise from the Δt-dependent N-R solver (Section 4.1.2). It also provides density-dependent metallicity and non-LTE molecular weights. Multi-redshift table interpolation tracks the evolving UV background (Haardt & Madau 2012) across cosmic time, eliminating the systematic bias of fixed-z tables.

(ii) **Jeans-independent holdback bypass.** The `jeans_bypass_holdback` option allows the Jeans criterion to operate ahead of the cosmological holdback, preventing artificial fragmentation and smoothing the star formation rate transition when levels are released. A configurable level limit (`n_jeans_bypass_levels`) and a physics-based density threshold (`d_jeans_thre`) provide complementary safeguards against runaway refinement cascades.

(iii) **Quantitative analysis.** We have shown that differences between the RAMSES and Grackle cooling rates propagate into 40–80 per cent errors in the Jeans length at low densities, and that the Δt-dependent N-R solver introduces level-dependent temperature biases that drive spurious refinement at AMR level boundaries. The exact integrator eliminates this entirely.

(iv) **Critical polytropic index.** The Jeans length in the polytropic regime scales as $\lambda_J \propto \rho^{(g_\star-2)/2}$. For $g_\star \geq 2$, λ_J is non-decreasing with density and the Jeans criterion self-terminates without any auxiliary mechanisms. However, increasing $g_\star$ from $5/3$ to 2 raises the temperature floor at the star-formation threshold by a factor of $\sim$4 (Table 5), necessitating a full recalibration of the star-formation parameters. The combination of $g_\star = 5/3$ with `d_jeans_thre` $\approx n_\star$ provides an effective alternative that preserves the existing calibration while limiting the cascade to 1–2 extra levels (Section 7).

(v) **Holdback diagnosis.** A direct numerical test at $z \approx 9$ demonstrates that with accurate, Δt-independent cooling, the cosmological holdback can be disabled without triggering a refinement cascade (Section 10.6). The holdback has historically been compensating for the level-dependent temperature artefacts of the N-R solver; with the exact integrator and multi-z tables, its role reduces to computational cost control. Turbulent pressure support, while physically important, cannot replace the holdback in the diffuse regime where it is most needed (Section 10.5).

(vi) **eEOS polytropic energy floor.** Disabling the holdback exposes a secondary failure mode: negative internal energy from the total-energy formulation in high-Mach-number post-feedback flows, which collapses the CFL time-step by $\sim$ 12 orders of magnitude (Figure **??**). We introduce a minimal energy floor $e_{\rm int} \geq C_{\rm eEOS}\,\rho^{g_\star}$ based on the existing star-formation polytropic equation of state (Section **??**). Unlike resolution-dependent pressure floors, the eEOS floor has no effect on diffuse gas thermodynamics ($T_{\rm floor} \ll T_{\rm eq}$ for $n \ll n_\star$) and only activates when a catastrophic numerical error would otherwise occur. Diagnostic tests confirm that the floor eliminates all instances of $e_{\rm int} < 0$ over 39 coarse steps past the original crash point, maintaining stable CFL time-steps (Table **??**).

Future work will extend the Jeans criterion to include turbulent pressure support (equation 43), using subgrid-scale turbulence models to estimate the unresolved velocity dispersion on the grid scale. A full star formation rate comparison from $z = 50$ to $z = 1$ between the standard cooling with holdback and the exact cooling without holdback is in progress. This turbulence-dependent refinement criterion, combined with the improved cooling module, holdback bypass, and eEOS energy floor presented here, will provide a comprehensive framework for accurate Jeans length resolution in next-generation cosmological simulations.

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